

RESEARCH ARTICLE

Computer Vision-Based Detection and Classification of Road Obstacles: Systematic Literature Review

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ABSTRACT Road obstacles are a major contributor to traffic accidents, making their accurate detection and classification vital for road safety and infrastructure maintenance. This paper presents a systematic literature review (SLR) of 95 peer-reviewed studies published between 2015 and 2024, specifically focusing on computer vision-based approaches for detecting and classifying road anomalies, including potholes, cracks, manhole covers, and speed bumps. Potholes were the most frequently studied obstacle (56 studies), followed by cracks (19), while manholes and speed bumps were the least represented (6 each). Detection tasks dominated (44.29%), with classification (30%) and combined approaches (25.75%) also explored. Deep learning techniques, particularly YOLO variants and CNNs, were the most used (87.14%), achieving state-of-the-art performance—e.g., YOLOv8 attained 100% accuracy and 99.50% mAP on benchmark datasets. Public datasets were used in 36.59% of cases, and 91.55% of studies relied on image-based inputs. Preprocessing techniques, such as data augmentation (30.86%) and annotation (24.69%), were prevalent. Although this review strictly focused on computer vision techniques, it identifies future research directions such as multi-sensor fusion, machine learning forecasting, crowdsourced data integration, and smart city infrastructure alignment. These broader approaches, though beyond this review's scope, are critical for developing more comprehensive, real-time, and context-aware road obstacle detection systems.

INDEX TERMS Intelligent transport systems, obstacle detection, obstacle classification, road safety, road features.

I. INTRODUCTION

An Intelligent Transportation System (ITS) [1] refers to the combination of intelligent infrastructure devices [2] interconnected to enhance road safety, improve traffic flow, and optimize transportation efficiency. ITS technologies integrate communication, sensors, and automation to ensure safe and efficient mobility. It was introduced in 1991, but it originated in the 60s of the 20th century when the United Kingdom government began investments in transport infrastructure and research [3]. ITS offers users a variety of road traffic applications. Moreover, it uses information and communication technologies to create new opportunities for businesses, open new markets, and boost economic activities.

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ITS power remains in real-time data collection and sharing, making decisions quicker. Thanks to this fact, it also has a great impact on the Internet of Things solutions, such as smart cities, by reducing traffic congestion [4]. Detecting and preventing accidents [5]. Lowering pollution and improving public transportation. Among the advantages of ITS is its ability to communicate with an Advanced driver-assistance system (ADAS) [6]. In real time, it sends real-time traffic updates.

The ADAS scope is vehicle-focused. It assists with the driver and safety features within the vehicles, which means that the primary users are the individual drivers and car manufacturers. It uses sensors (cameras, Radars, LiDars) and onboard computers as key technologies, and its main goal is to assist the drivers in real-time and improve vehicle safety. In short, ADAS is not exclusively an ITS application. Still,

it can be considered one of its key applications, particularly in the context of Vehicle-to-Everything (V2X) [7] communication and smart mobility [8].

As transportation solutions and technologies are continuously growing, ADAS and ITS are increasingly being integrated as well as Artificial Intelligence to provide smart, efficient, and safer transportation networks. AI relates to the other two components strongly. From an ADAS perspective, AI in Sensors and Perception ADAS relies heavily on sensors, where AI algorithms analyze the collected data to make real-time decisions. For example, AI can help ADAS systems recognize pedestrians [9], traffic signs [10], and other vehicles in complex environments. Improved Decision-Making [11]: AI enables ADAS to learn from previous experiences, improving its ability to predict hazards and assist the driver. From the ITS perspective, AI provides a big enhancement to the ITS. AI can analyze real-time traffic data from road sensors and adjust traffic signals automatically to reduce congestion [12] or optimize flow [13] based on demand. AI can quickly detect accidents or unusual traffic patterns and alert drivers or traffic control centers. Finally, the combination of ADAS, ITS, and AI concludes multiple benefits, such as Enhanced Safety: AI enhances ADAS systems by enabling them to better understand and react to the environment. When combined with ITS, this creates a safer driving environment for all road users and improved Traffic Flow. ITS with AI optimizes traffic management in real-time, helping to reduce congestion and improving overall travel efficiency. Autonomous Vehicles [14]: AI, ADAS, and ITS play a critical role in the development of autonomous vehicles, where ADAS provides on-board safety features, ITS offers external communication and data, and AI enables decision-making and learning. Environmental Benefits: AI-driven traffic management can optimize routes, reducing fuel consumption [15] and emissions by preventing congestion and idle times. ADAS, ITS, and AI are key components of the future of transportation. By combining these technologies, transportation networks will be safer, more efficient, and smarter.

This systematic literature review aims to provide a comprehensive analysis of the current body of work on computer vision-based approaches, particularly Deep Learning-driven Convolutional Neural Network (CNN) [16] Models for the detection and classification of road obstacles, and to highlight potential areas for contribution. To ensure efficient analysis, we adopt the Systematic Literature Review, a methodological approach that aims to provide an exhaustive and impartial synthesis of available knowledge on a given subject. By following a predefined protocol, SLR helps minimize selection bias and provides a clear, comprehensive overview of previous research. This paper covers research published between 2017 and 2024 and presents a detailed analysis of previous work. It covers key steps, including identification of research questions, selection of relevant studies, data extraction, and model training and analysis.

The contributions of this research are outlined as follows:

- A focused review of recent academic research on computer vision-based detection and classification of road obstacles.
- An in-depth assessment of data sources, datasets, features, and preprocessing techniques, including feature extraction and selection methods.
- A thorough evaluation of methodologies and models applied in obstacle detection studies, along with their reported performance outcomes.

The findings of this review have practical implications for the deployment of road obstacle detection systems in intelligent transportation solutions. Specifically, the reviewed techniques can support autonomous vehicle navigation, urban infrastructure monitoring, and smart city development by enabling accurate and real-time detection of road hazards.

Our paper is organized as follows. In Section II, we provide a literature review of works related to our research context, followed by Section III, which details the research methodology we employed to select and analyze the papers included in this study. Section IV presents a comprehensive overview of road obstacle detection and highlights the main results of our work. In Section V, we summarize the findings, while Section VI offers a broader discussion of the overall study. Finally, we conclude the paper with Section 7, which provides the main conclusions and insights.

II. RELATED WORKS

This section presents an overview of research advances in the field of road obstacle detection, including the detection of defects and road features such as potholes, cracks, bumps, and manholes. It explores computer vision-based techniques and approaches used to identify these anomalies, which affect road quality and user safety.

To provide a clear, structured overview of existing research into road obstacles, this section is divided into three distinct parts. The first part focuses on the detection of defects and road features, exploring the different methods and technologies used. The second part deals with the classification of road defects by analyzing approaches for categorizing these obstacles according to their characteristics. The third part examines work that integrates both the detection and classification of road defects and features. Finally, the related works section concludes with a synthesis, where we summarize the key findings and insights from the discussed studies.

A. DEFECTS AND ROAD FEATURES

1) DETECTION OF DEFECTS AND ROAD FEATURES

The study [17] Aims to detect potholes in real-time using YOLOv3 applied to webcam video streams, with visual outputs mapped via Google Maps API. The model is trained on 330 images over 2000 epochs with a learning rate of 0.001, achieving 0.90% precision, 0.41% recall, and 0.65% mAP. The authors suggest using more powerful GPUs and expanding the dataset for improvement.

Building upon vision-based techniques, the study [18] proposes enhancing road safety through RGB segmentation and Otsu's thresholding on over 80 images processed in MATLAB. The model achieves 82% accuracy and an average execution time of 465.71 ms per image. Extending detection capabilities to autonomous vehicles, the study [19] uses YOLOv8 trained on 665 annotated images. It demonstrates perfect performance (100% precision, recall, and F1-score) after 200 epochs on Google Colab. Similarly, addressing autonomous driving, the paper [20] applies CNNs to images captured via a Raspberry Pi camera. The model, trained on a test track, attains 88.6% accuracy but encounters challenges during tight turns. Broadening model comparisons, study [21] evaluates YOLOv2 (standard and enhanced) and Faster R-CNN (with VGG16 and ResNet-18) on 1300 annotated images. The enhanced YOLOv2 yields the highest performance with 87% precision and 89% recall.

Shifting to sensor-based input, research [22] Transforms smartphone vibration data into pixel images to train CNNs. The resulting model reaches up to 93.24% validation accuracy and detects potholes with 80–90% accuracy.

In the domain of road obstacles, the article [23] Introduces a MobileNet SSD-based approach for detecting speed bumps. Using 3,000 labeled images, it achieves over 97% accuracy and estimates obstacle distances within ± 20 cm.

To explore CNN-based automation, study [24] Compares a custom 7-layer CNN (trained on 1,440 images) with InceptionV3 using transfer learning. While both perform well, InceptionV3 offers slightly better generalization but is more sensitive to noise.

Optimizing for embedded deployment, paper [25] Employs knowledge distillation to compress a complex model (TM) into a simpler one (SM) using a dataset of 1,459 images from Nigeria. Results are pending, but the study emphasizes model efficiency.

In a regional context, the article [26] Targets South African road conditions using YOLOv3, SSD, and Faster R-CNN on 1,910 images. YOLOv3 proves most effective for real-time detection with a 34.7% precision and 42% F1-score.

Continuing the evaluation of YOLO variants, the study [27] uses vehicle-mounted cameras and compares YOLOv3, Tiny, and SPP versions. YOLOv3 SPP achieves the highest mAP at 88.93%.

Expanding to dimensional analysis, in the article [28] Combines YOLOv3/YOLOv4 with triangular similarity to estimate pothole size. YOLOv4 outperforms YOLOv3 with mAP of 0.933 and IoU of 0.741. Integrating traditional sensors, research [29] Proposes a CNN-DL model based on accelerometer and ultrasonic data. Simulated in MATLAB, the system attains 96.38% accuracy and outperforms classical methods. For lightweight mobile applications, study [30] Applies YOLOX-nano to detect diverse pothole shapes. Achieving an AP of 85.6%, it surpasses YOLOv5m while maintaining a compact 7.2 MB size.

Comparative analysis continues in the article [31], which benchmarks YOLO, Faster R-CNN (ResNet50), and MVGG16. Faster R-CNN achieves 91.9% accuracy, while YOLO offers better speed, and MVGG16 balances both. The article [32] Corroborates these findings, with an emphasis on improved detection under extreme weather using stereo camera-based depth estimation.

In the Indian context, the study [33] Applies F-RCNN with Inception-V2 via TensorFlow API on 1,500 images. Retraining final layers boosts performance, enabling effective real-time detection.

Focusing on adverse weather, the study [34] Compares YOLOv3–v7 and several R-CNNs. YOLOv7 performs best during the day, while R-CNNs excel under low-light conditions.

The study [35] Confirms YOLOv3's real-time potential on 1,200 images from Port Harcourt, reaching 97% accuracy during live video detection. Leveraging data augmentation, in a study [36] Trains YOLOv5 on a custom dataset, achieving a mAP of 99.90% and improved model generalization.

Using generative methods, study [37] Applies a GAN model for speed bump segmentation. With Drive data, it achieves 93.8% IoU and 0.9840 SSIM. In infrastructure mapping, the study [38] Evaluate RetinaNet (ResNet-50/101) on 297 images from Brazil. The model achieves mAP > 90% and surpasses Faster R-CNN in detecting sewer gaps.

Shifting to model lightness, study [39] Utilizes Tiny YOLOv4 on Kaggle images, achieving an AP of 0.734 and mAP of 0.458 for pothole dimension estimation. In addition, the study [40] YOLOv2, YOLOv3, and YOLOv3 Tiny are trained on 1,500 Indian images. YOLOv3 Tiny performs best with an mAP of 72.12% and an accuracy of 76%.

Expanding to edge devices, study [41] benchmarks YOLOv1–5, Tiny-YOLOv4, and SSD-MobileNetv2. YOLOv5 achieves 95% accuracy, while Tiny-YOLOv4 provides better real-time balance (90% at 31.76 FPS).

In study [42] Research compares R-CNN and SSD MobileNet on Bangladeshi road images. Accuracy reaches 93%, but lighting and video quality affect performance. Targeting structural damage, study [43] upgrades YOLOv3 with MobileNetV2 and CBAM. It achieves 89.16% accuracy, 91.16% recall, and 20.56 FPS. Finally, study [44] Develops a real-time Raspberry Pi-compatible system using YOLOv5m. While mAP reaches 76%, performance is limited to 2–3 FPS, suggesting a need for optimization.

2) CLASSIFICATION OF DEFECTS AND ROAD FEATURES

The study [45] Aims to classify images into speed bumps, potholes, or normal roads using CNN models. With a dataset of 714 images (179 speed bumps, 231 potholes, and 304 normal roads), the best-performing model achieved 98.13% accuracy, demonstrating an efficient and lightweight architecture for road anomaly classification. Similarly, study [46] Proposes an advanced CNN-based method for pothole classification to improve autonomous vehicle safety. Trained

on both public and field-captured data, the model achieves 99.02% accuracy, 99.03% sensitivity, and an F1-score of 98.33%, outperforming many existing approaches.

Expanding the modality, the article [47] Uses thermal images captured with FLIR ONE cameras, comparing self-built CNNs and ResNet models. ResNet101 delivers the highest validation accuracy of 97.08%, highlighting the promise of thermal imagery combined with deep learning.

Addressing the issue of manual road inspections, a study [48] Utilizes pre-trained models (ResNet50, ResNet152, InceptionV3, DenseNet201) to classify images from a dataset of 722 training and 116 test images. Inception-ResNetV2 and DenseNet201 both achieve 89.66% accuracy, suggesting viable alternatives to traditional road grading.

With a similar goal, study [49] applies CNN-based classifiers (ResNet50, InceptionResNetV2, VGG19) to online and Kaggle images. VGG19 achieves outstanding performance, reaching 97% accuracy on highways and 98% on muddy roads.

A more sophisticated pipeline is presented in the study. [50], which employs ResNet50 for feature extraction, bAMDTO for optimization, and random forest for classification. Using SMOTE for class balance, the model reaches 99.8% accuracy and a 99.05% F1-score.

In the context of smart cities, research [51] Explores lightweight CNN models (4-layer CNN) compared to MobileNetV2, InceptionV3, and NASNet. Trained on 6,000 images under varied conditions, the model achieves up to 98% fold accuracy and 97.47% overall with only 8.4M parameters. For embedded AI applications, study [52] Fine-tunes pre-trained ResNet-18 and VGG-11 models for classifying potholes, bumps, and cracks. The adapted models achieve nearly 100% across all metrics, proving their reliability for autonomous vehicles. Focusing on cracks, paper [53] Combines CNN classification with adaptive image segmentation. Trained on 40,000 balanced images, the model reaches 99.92% classification and 98.70% segmentation accuracy. Study [54] Tackles pothole classification on bituminous roads using VGGNet with transfer learning. The model reaches 96% validation accuracy and shows potential for expanding to broader defect detection and road quality assessment.

The study [55] Uses an Android app to collect sensor data (accelerometer, gyroscope, speed) during donkey rides. CNN models outperform LSTM in classifying road conditions, achieving 99% accuracy for normal roads and 92% for speed bumps. Likewise, study [16] Uses 1D-CNNs to classify potholes based on accelerometer data from smartphones. Achieving up to 96.29% accuracy and an F1-score of 91%, this method provides an affordable solution without specialized equipment.

To address pothole classification on South African roads, a study [56] Develops a CNN trained on preprocessed images from multiple sources. The model achieves 99.80% on training data and 92.72% on testing, with suggestions to improve accuracy via higher-quality images.

In developing country contexts, the study [57] Targets automated pothole detection for Timor-Leste. Using CNNs trained on images under varied lighting and terrain, the model achieves 99.80% accuracy, 100% precision, and 99.60% F1-score, with plans for smartphone-based systems. With a similar objective, study [58] Classifies potholes using CNN on video-derived images. After pre-processing and augmentation, the model achieves 99.80% accuracy, 99.60% recall, and an F1-score. Study [59] Shifts focus to crack detection using a CNN trained on 3,000 images. After segmentation and thresholding, the model achieves 96.20% training accuracy, 96.50% validation, and 94.50% test accuracy.

In a post-disaster context, study [60] uses ResNet152 features combined with SVM, KNN, AdaBoost, and ensemble learning [61] To classify road cracks. Tested on the KERCD-2023 dataset, the SVM-ensemble model reaches 98.68% accuracy, aiding emergency response.

Improving detection precision, study [62] applies transfer learning [63] With ResNet50 to classify pothole images. The model achieves 98% accuracy, with future work proposed for severity estimation and repair prioritization.

In study [64] ResNet50 and VGG16 are used to classify 1,280 images of potholes and smooth roads. ResNet50 outperforms with 98.66% accuracy versus 96.31% for VGG16.

Article [65] Introduces a DenseNet121-based classifier achieving 99.3% accuracy, outperforming ResNet50 and VGG19. The model offers an efficient alternative for road condition monitoring.

Lastly, study [66] Optimizes VGG19 using a two-layer filter reduction algorithm for drone-captured road images. The optimized model retains an F1-score of 96.4% while reducing filter count by 52%, outperforming AOA and PSO methods. This lightweight solution offers strong potential for automated road inspections using aerial imagery.

3) DETECTION AND CLASSIFICATION OF DEFECTS AND ROAD FEATURES

Unlike the previous sections, which address papers focusing on either detection or classification of reading defects and road features separately, this section covers papers that focus on both tasks. These papers present approaches that simultaneously detect and classify road defects and features.

The study [67] Aims to detect and classify potholes and speed bumps on Indian roads using a combined approach with CNN (ResNet-50) and YOLO. The model classifies road surfaces into three categories (normal, pothole, speed bump) and localizes them with YOLO. Trained in MATLAB, the system achieves 88.9% classification accuracy and performs well in identifying bumpers, though data scarcity under various conditions remains a challenge. Following this is another publication. Reference [68] presents an enhanced CNN model for UAV-based road inspection, targeting early detection and classification of pavement cracks, potholes, and lane markings. The CNN with 13 convolution layers and the Mish activation function achieves an average accuracy of 94%.

Integration with YOLO enables real-time detection, demonstrating an effective solution for aerial road inspections.

In paper [69], researchers introduce YOLOv5s-M to detect alligator cracks, incorporating a large-scale detection layer, a generalized FPN module, improved IoU loss, and a decoupled sensing head. Tested in Beijing's Fengtai district, it achieves 78.2% accuracy, 72.1% recall, and 75% F1-score, outperforming previous models. In [70], leverage WorldView-3 satellite imagery and deep CNNs to detect and classify anomalies such as potholes and landslides. Images are preprocessed with Nested U-Net and classified using Deeplab. While effective, the approach faces limitations in managing environmental interference. Using Xilinx Edge AI platforms, study [71] combines YOLOv3 with dual cameras of varying FoV to detect pavement defects. Trained on the TTPDID dataset (62,000 samples), it achieves 71% pothole detection accuracy and 27.8 FPS on FPGA, showing potential for deployment on embedded systems.

For research [72], The authors introduce MGB-YOLO, a YOLOv5s model enhanced with MobileNet-V3, GAM, and BottleneckCSP for detecting and classifying pavement overhangs using stereo depth cameras. With a dataset of 1,000 augmented images, the model achieves 96.6% accuracy, outperforming Faster R-CNN and SSD. The article [73], use of data from smartphone sensors is analyzed with machine learning algorithms to detect anomalies. SVM achieves the highest accuracy with 88.55%, followed by decision trees with 88.35%, and MLP neural networks with 79.86%.

There is another [74] Evaluates several models—YOLOv3, SSD, HOG+SVM, and Faster R-CNN—for pothole detection/classification. YOLOv3 offers the best balance between speed and accuracy at 82% accuracy, making it the preferred choice. On the other hand, the study [75] Focuses on automatic detection and classification of road cracks using YOLOv4. Trained on 2,000 labeled images and supported by NVIDIA RTX 3050 Ti, the model achieves 80% accuracy, 81% IoU, and 77% F1-score.

Optimizing YOLOv5s, the research [76] Replaces its backbone with MobileNetV3, integrates KMeans for anchor box selection, and adds CA attention. Evaluated on RDD2020, the model achieves 57.2% accuracy, 56% recall, and 53.6% mAP. In addition, the article [77] Aims enhances CNN with attention fusion and salient feature normalization modules (NAM), evaluated on RDD2020 and CNRDD. The model achieves F1-scores of 51.72% and 34.73%, respectively, validating the use of NAM and attention mechanisms.

In [78] Images captured in Dhaka are preprocessed and analyzed using R-CNN, Faster R-CNN, and SVM. Faster R-CNN yields superior accuracy with 98.02% on training and 99.80% on validation, with minimal loss. The study [79] Explores ensemble learning (EP, EM, EM+EP) with ultralytics-YOLO on 21,041 images from the 2020 IEEE Big Data Challenge. The hybrid ensemble approach achieves an F1-score of 0.67, enhancing robustness despite longer inference times.

For real-world deployment, study [80] Uses smartphone-based images and Faster R-CNN with ResNet-152 for eight defect types. The model improves upon its ResNet-101 variant with an average F1-score of 0.6255. In addition, research [81] Applies a CNN to classify defects from images and videos. With a 6,000-image dataset (4,500 training, 1,500 testing), the model achieves 94% accuracy, 92% precision, and 96% recall. In Rwanda, paper [82] Deploys a YOLOv5-based system using a smartphone to detect road damage in real-time. Trained on 3,039 images, the model achieves a mAP of 0.85 and integrates GPS tagging through a dashboard interface.

Finally, study [83] Develops a crack classification system using SSD and a receptive field module (RFB) to improve feature extraction. Trained on 4,000 images with 1,200 refinement images, the model achieves an F1-score of 0.7787, with future work focusing on data enrichment and better feature fusion.

B. SYNTHESIS

The state of the art on road fault detection and improvements shows a diversity of methodologies and approaches to improve road safety and the efficiency of autonomous transport systems. Among the obstacles most often addressed in research are potholes, cracks, bumps, and other road defects that may pose hazards to vehicles. For work on road obstacle detection, various deep learning models were used. The YOLO (You Only Look Once) and Faster R-CNN algorithms are particularly popular for their ability to perform high-precision real-time detections. YOLO is distinguished by its speed, processing the entire image in a single step to locate objects simultaneously. Faster R-CNN, meanwhile, uses a network of region proposals (RPNs) to suggest areas of interest before classifying them, offering excellent accuracy but often at a higher computational cost. Other models, such as SSD (Single Shot MultiBox Detector) and CNN, are also commonly used for obstacle detection. SSD, similar to YOLO, allows one-step predictions using a grid of predictions at different scales to detect objects. CNNs, on the other hand, are a fundamental architecture in many detection approaches because of their ability to extract relevant features from images. Furthermore, for research aimed at the classification of obstacles, convolutional neural networks (CNNs) are most effective because they can distinguish between different types of obstacles based on specific visual characteristics. Advanced models such as ResNet, VGG, and Inception are frequently used to improve classification accuracy. ResNet is known for its residual layers that facilitate the drive of very deep networks, VGG is appreciated for its simplicity and ability to capture detailed features, whereas Inception uses different-sized convolution modules to better understand variations in images. For approaches combining both detection and classification, YOLO and Faster R-CNN are often preferred because of their effectiveness in both tasks. These models are capable of detecting and classifying

obstacles in a single pass, making them highly suitable for real-time applications. Cascade R-CNN and SSD are also used, offering robust solutions for simultaneous detection and classification through their specific architecture and their management of different object scales. On the other hand, image preprocessing techniques such as data augmentation, resizing, and annotation are used to improve the performance of these models. Some research also incorporates complementary data such as vibration sensors and GPS data to enrich visual information and enhance the reliability of obstacle detection systems. In addition, most studies in this area rely on private data sets, although some also use public data sets. These data sets are mainly composed of images captured by cameras mounted on vehicles or smartphones. The work analyzed highlights the importance of preprocessing techniques, including data enhancement to improve model performance. This technique helps to diversify the training data, which strengthens the models' ability to handle a wider range of real-world scenarios. However, future research should focus on improving adverse detection, extending data sets, and integrating these systems into intelligent transport infrastructures. The various studies show a constant evolution towards more precise, fast, and adaptable road obstacle detection systems in different contexts, contributing to better road safety and optimized infrastructure management.

III. SEARCH METHODOLOGY

The research methodology is a key component that guides the entire investigation process. It defines the steps and methods used to collect, analyze, and interpret data. In this section, we will outline the research questions that guide our study, thus providing a clear basis for exploration.

A. RESEARCH QUESTIONS

The research methodology is developed in response to specific questions as follows:

- RQ1: What types of road obstacles are covered in the literature?
- RQ2: What are the objectives of studies covered in the articles?
- RQ3: What data sets and data sources are used?
- RQ4: What are the preprocessing techniques applied to the data?
- RQ5: What techniques and models are used in the literature?
- RQ6: What is the performance of the models used in the articles studied?

B. SEARCH STRATEGY

To ensure a comprehensive and relevant search of existing literature, a rigorous selection of keywords was carried out. The selected keywords were carefully chosen to cover a wide range of topics related to the detection and classification of road hazards and damage. These keywords include "Obstacle Detection for Autonomous Vehicles", "Pothole Detection",

"Speed bump Detection", "Deep Learning based Road Damage Detection", "Manhole cover Detection", and "Road Obstacle Detection using Deep Learning". Each term has been carefully selected to encompass studies on innovative methods such as computer vision, sensing sensors, image processing techniques, as well as the use of specific databases for road damage research. This approach allows for a comprehensive exploration of existing research, identifying technological advances and best practices in the field of road infrastructure monitoring and maintenance

C. RESEARCH DATABASE

To ensure a comprehensive collection of relevant studies, several academic databases were explored. The key sources used in this systematic review include:

- IEEE Xplore Digital Library: Managed by IEEE, it offers articles, conference papers, standards, and books in engineering and computer science.
- Scopus: A multidisciplinary database by Elsevier, covering science, technology, medicine, social sciences, and arts.
- Google Scholar: A search engine for academic literature, including journal articles, theses, and conference papers across various fields.
- SpringerLink: Operated by Springer Nature, providing access to publications in science, technology, and medicine.

The combined use of these databases ensures thorough coverage and critical evaluation of the available knowledge in the field studied

D. STUDY CHOICES

Before undertaking this analysis, rigorous inclusion and exclusion criteria were established. These criteria ensure that only the most relevant and high-quality studies are selected for in-depth analysis. Inclusion criteria determine the specific characteristics that articles must meet to be considered, while exclusion criteria identify publications that do not meet established standards. These methodological approaches not only ensure consistency and rigor but also the reliability of the conclusions that will be drawn. The criteria were applied systematically to guide research and the selection of the most relevant studies in the field. Table 1 details the inclusion and exclusion criteria adopted for this systematic review:

This systematic review on road obstacle detection and classification follows the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. PRISMA provides a standardized framework to enhance transparency, quality, and reproducibility in systematic reviews. It includes a checklist and flowchart to guide study planning, selection, and synthesis. By adopting PRISMA, this study ensures a rigorous methodology, improving the credibility and thoroughness of the results. The process is detailed in a figure illustrating the PRISMA scheme's steps.

TABLE 1. Inclusion and exclusion criteria.

Criteria	Inclusion Criteria	Exclusion criteria
Publication period	Papers published between 2017 and early 2024 were considered. This period allows coverage of recent and relevant developments in the field.	Articles published before 2017 were excluded to focus on the most recent research in the field.
Publication type	Eligible research includes articles published in journals and conference proceedings.	Research not published in peer-reviewed journals or conference proceedings was excluded to maintain a high level of scientific quality and rigor.
Academic sources	Studies must come from leading academic sources such as IEEE Xplore, Scopus, SpringerLink, and Google Scholar.	Studies from non-recognized sources such as IEEE Xplore, Scopus, SpringerLink, and Google Scholar were excluded to ensure access to reliable research.
Language	Articles included must be written in English	Articles written in a language other than English have been excluded
Techniques used	Articles must use ML or DL algorithms and use computer vision-based techniques.	Studies not using ML, DL, or computer vision techniques were excluded.
Topic	The selected studies should focus specifically on the detection and classification of road obstacles.	Studies not specifically focusing on road obstacle detection and classification were excluded.

The PRISMA diagram illustrates the article selection process for the systematic literature review (SLR). It begins with the identification step, where a search across various databases yields 280 articles. After removing duplicates and applying exclusion criteria during the selection step, 185 articles remain. These are then screened by title, keywords, summary, and conclusion, excluding 72 more, leaving 113 articles. Finally, in the eligibility stage, full-text reviews excluded 43 additional articles, resulting in 70 articles included in the SLR.

E. DATA ANALYSIS

Data analysis is a critical step in understanding the scope and impact of road barrier research. This section examines the articles included in our study in a general way, quantifying the number of publications in newspapers and conferences. Table 2 presents the different types of sources of studies examined.

To better visualize the distribution of publications by source type, the following Fig. shows a bar chart and a pie chart, which provide a clear and concise overview of the distribution of studies between journals (e.g., Journal of the Year). Concise overview of the distribution of studies

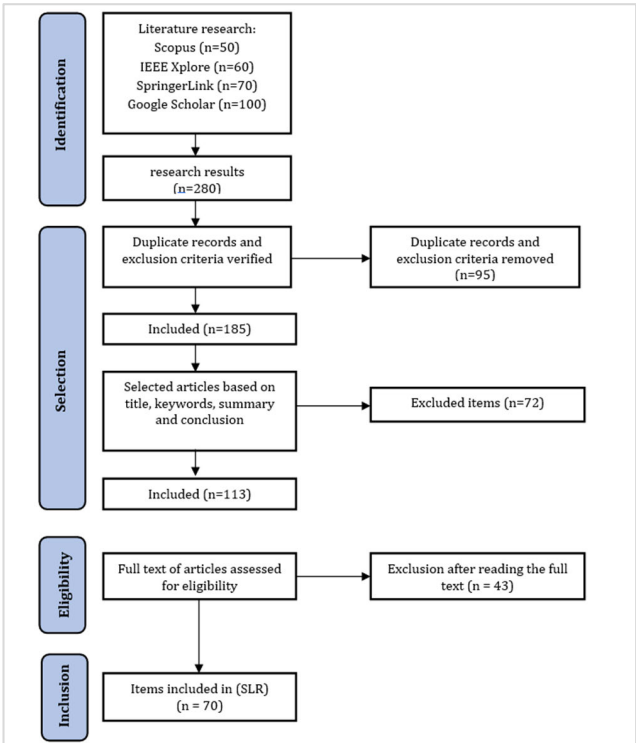


FIGURE 1. PRISMA diagram.

TABLE 2. Type of source for selected studies.

Source Type	REFERENCES
Conference Proceedings	[18] [80] [57] [70] [53] [32] [67] [27] [29] [21] [20] [42] [65] [33] [39] [43] [28] [83] [75] [17] [78] [49] [23] [74] [79] [62] [64] [37] [40]
Journal Article	[41] [56] [45] [71] [34] [77] [30] [52] [44] [16] [51] [84] [68] [44] [19] [24] [50] [66] [73] [58] [36] [69] [81] [35] [48] [59] [47] [26] [85] [54] [55] [72] [76] [86] [31] [22] [38] [25] [60] [46]
Report	[82]

between journals (Journal articles) and conferences (Conference Proceedings), and reports (Report).

Fig. 2 shows two graphs illustrating the distribution of publications by type of source. The bar chart shows that of the publications studied, 40 are journal articles, 29 are conference proceedings, and 1 is a report. This highlights the predominance of journal articles, followed by conference proceedings, with a conference proceeding, with a very marginal presence of reports. The pie chart sector offers a complementary perspective by representing this data in percentage terms.

Journal articles make up 57.14% of total publications, conference proceedings 41.43%, and reports 1.43%. The Fig. reveals that journal articles are the main source of research dissemination, reflecting a preference for peer-reviewed and often more detailed publications. More detailed. Conference proceedings, although slightly less

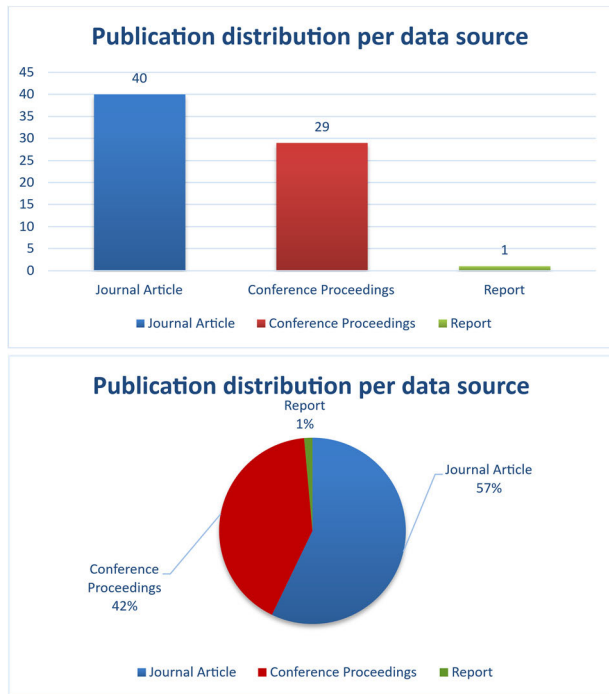


FIGURE 2. Breakdown of publications by type of source.

numerous, also play a significant role, indicating a preference for more detailed, peer-reviewed publications, also play a significant role, indicating the importance of conferences as platforms for sharing recent work and obtaining immediate feedback. To complete our analysis, we will now examine the distribution of articles by year of publication. This temporal perspective will enable us to understand the trends and evolution of road barrier research over time. The following Fig. below illustrates the annual distribution of selected publications.

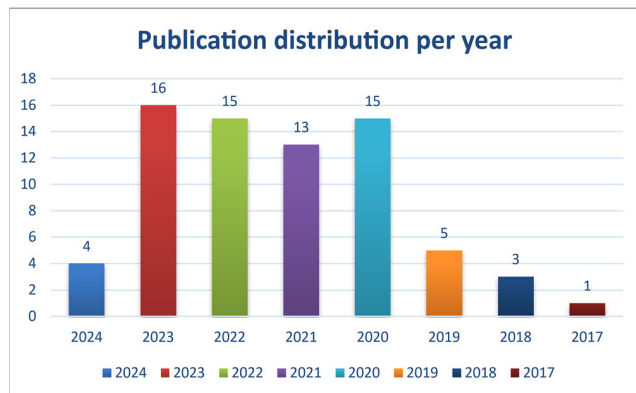


FIGURE 3. Annual distribution of publications.

Fig. 3 shows that most articles were published in 2021 and 2023. This high concentration of publications in these two years suggests a particular interest or significance in the field during this period.

IV. ROAD OBSTACLE DETECTION: A COMPREHENSIVE OVERVIEW

A. TYPES OF ROAD OBSTACLES DESCRIBED IN THE LITERATURE (RQ1)

As part of our literature review, it is important to know the different types of road obstacles researchers discuss. Road obstacles represent significant challenges for road safety and infrastructure maintenance. Identifying these is not only to guide research efforts but also to improve detection and prevention strategies. Detection and prevention strategies. The Fig. below examines the various types mentioned in the literature, providing an overview of current trends and research trends and research priorities:

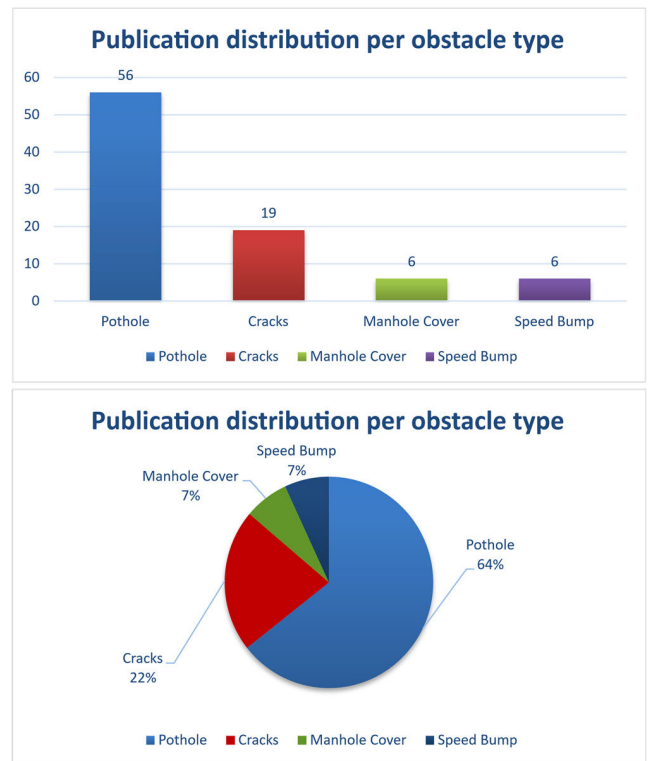


FIGURE 4. Types of obstacles discussed in the literature.

Fig. 4 shows two types of graphs illustrating the distribution of obstacle types covered in the literature. The bar graph clearly shows the distribution of the different types of obstacles. Potholes dominate with 56 occurrences, representing a substantial proportion of total interventions. Cracks are the second most frequent type of obstacle, with 19 documented cases. Speed bumps and manhole covers are the least frequent, with 6 occurrences each. These results highlight the prevalence of potholes as the main type of obstacle requiring intervention. This predominance is mainly due to the significant damage that potholes can inflict on vehicles, resulting in costly repairs and increased safety risks for drivers. Cracks, although less frequent, also account for a significant proportion of interventions, suggesting areas requiring regular monitoring and maintenance.

Current literature focuses mainly on the obstacles with the greatest impact on road safety, road safety. This trend reflects the importance of these obstacles in research and the need to develop effective technologies and strategies for their detection and management.

B. THE OBJECTIVES OF STUDIES COVERED IN THE ARTICLES (RQ2)

Before presenting the statistical results, we outline our approach to categorizing studies based on the application goals of ML or DL techniques: detection, classification, or both. Detection identifies the presence and location of road features, classification assigns them to specific categories, and combined approaches address both tasks. This categorization helps evaluate methodologies and their effectiveness. This section of our analysis explores the different objectives of the studies reviewed in the literature. The following Fig. provides a detailed visualization of the specific objectives pursued by the researchers, providing an overview of the various priorities and orientations in this field.

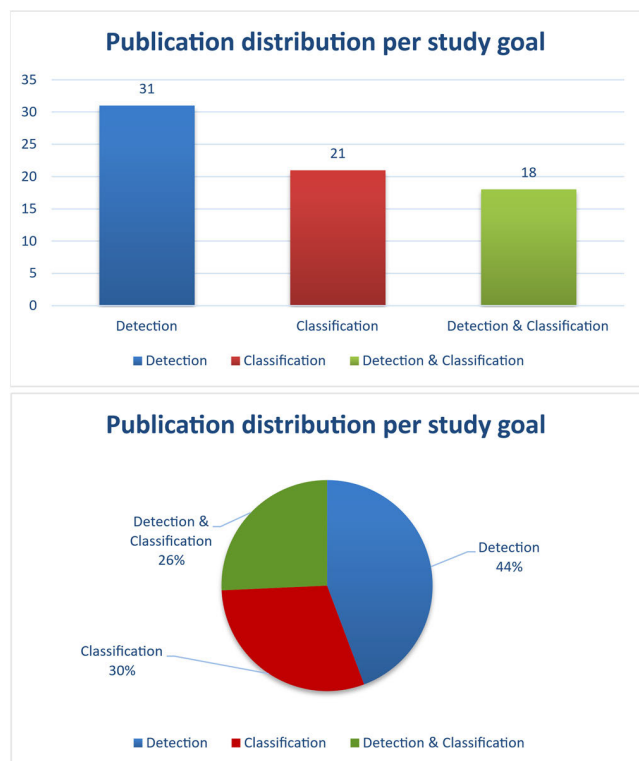


FIGURE 5. Types of studies covered in the literature.

Fig. 5 shows that studies focusing on detection are the most frequent, with 31 articles (44.29%), representing the majority of research in this field. Classification studies follow with 21 articles (30%), showing a significant but less interest compared with detection alone. Finally, studies combining both detection and classification accounted for 18 articles (25.75%), highlighting an integrative approach in a substantial portion of the research.

These results highlight several key points. Firstly, the predominance of detection studies suggests the identification of specific objects or phenomena specific objects or phenomena. This focus could be due to the complexity and importance of accurate detection in various application contexts. Secondly, the significant number of classification studies indicates a focus on the organization and interpretation of data once detected, which is obvious for more advanced applications and in-depth analyses. Combined detection and classification and classification, although fewer in number, reveal a tendency to integrate the two processes for a more comprehensive and efficient approach. This integration could be indicative of the growing maturity in the research field, where hybrid solutions are being explored to overcome the limitations of singular approaches.

Fig. 6 illustrates a bar chart showing the distribution of papers based on their goal and the type of obstacle they focus on, categorizing 70 studies based on their goal (Classification, Detection, or Detection & Classification) and the type of road obstacle being studied. In classification studies, normal roads (19 studies), potholes (17 studies), and cracks (5 studies) are most common. Detection studies focus heavily on potholes (26 studies), with other obstacles like manhole covers (2 studies) and cracks (1 study). Detection & classification studies cover a range of obstacles, with cracks (13 studies), potholes (13 studies), and manhole covers (4 studies) being prominent. The Fig. highlights that potholes are the most frequently studied obstacle overall, particularly in detection studies, while normal roads are the primary focus in classification studies.

C. DATA SETS AND DATA SOURCES USED (RQ3)

Datasets play a crucial role in the ML process; the models can be trained, validated, and tested on these datasets, ensuring the ability to generalize to new scenarios. The available datasets are essential to cover a wide range of use cases in the field of computer vision, particularly those involving camera-based systems, ranging from urban roads to rural environments, and to account for variations in weather conditions, road surfaces, and obstacles. These datasets can be categorized into two main types: public and private. Public datasets, often accessible to the research community, facilitate model performance comparison and encourage collaborative innovation. In contrast, private datasets, typically specific to organizations or projects, offer specialized data that addresses particular needs but may limit their broader applicability. To conduct a comprehensive and representative analysis, we distinguish between these two types of datasets and handle them separately, recognizing their respective contributions to the overall machine-learning process. Fig. 7 illustrates the distribution of the dataset by availability of the dataset, in order to have a clearer image of the available dataset introduced in the related works section.

Out of the total datasets, 52 are private, meaning they are not publicly accessible or require special permissions to access. In contrast, 30 datasets are public, meaning they are freely available for use and can be accessed by anyone. This

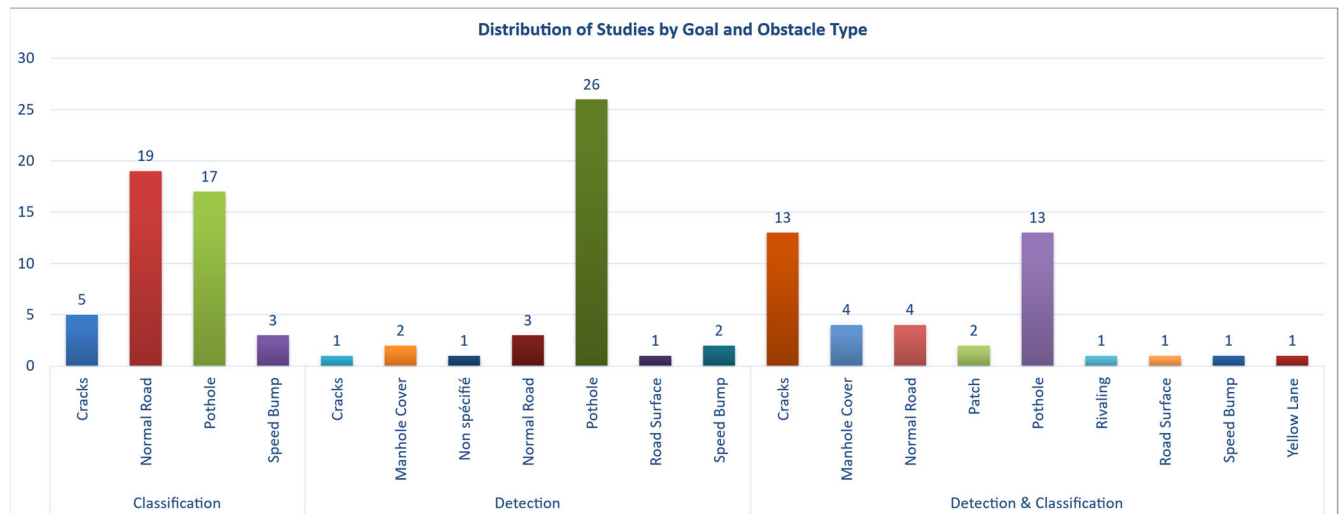


FIGURE 6. Distribution of studies by goal and obstacle type.

suggests that the majority of the datasets are private, while a smaller portion is openly accessible to the public. The next section will focus on the analysis and discussion of only the public datasets.

1) PUBLIC DATA SETS USED

Public data sets are a valuable resource for research. Their accessibility facilitates the comparison and validation of results on a wider scale. To reduce redundancy and avoid detailing minor variations, we grouped the obstacles as follows (P: Pothole, C: Crack, B: Bump, R: Rutting, M: Manhole Cover, NR: Normal Road, and S: Storm Drain). Table 3 explores the different public data sets used in the literature for obstacle detection, including their reference, size, type, and classes:

To gain a better understanding of the most influential public datasets in research on road obstacles. The following Fig. illustrates the distribution type of obstacles per dataset.

Fig. 8 presents a comparative analysis of the types of road obstacles existing in the public data sets used. Obstacles are classified into four main categories: potholes, cracks, speed bumps, and manhole covers. We can see that potholes are the most frequently encountered obstacles in most data sets. For example, the “Pothole600”, “Pothole Image Dataset,” and “Pothole Detection Dataset” focus exclusively on potholes. Cracks are also well represented, notably in the “Road Cracks526” and “Concrete Crack “Concrete Crack Images for Classification”. Speed bumps, though less frequent, appear in sets such as “Speed Bumps”, appear in sets such as “Speed Bump and Pothole Dataset”. Manholes are the least represented, appearing mainly in the “Storm drain and manhole detection” dataset. This Fig. highlights potential gaps in existing datasets and identifies the types of obstacles that could benefit from further may require further attention in future research.

Fig. 9 shows that the most used datasets are the Pothole Detection Dataset and the Annotated Potholes Image

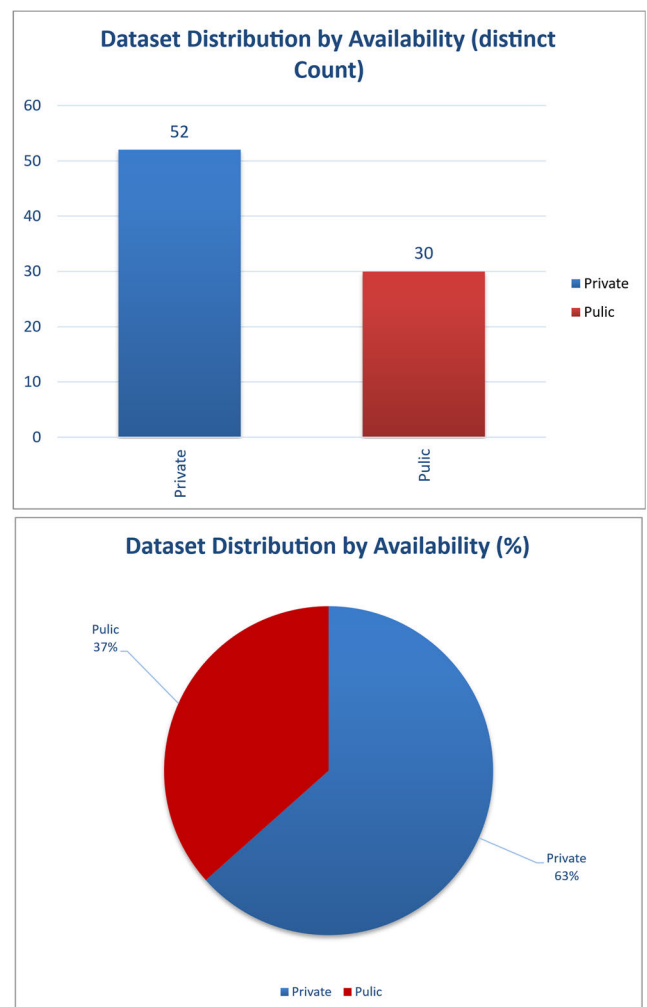


FIGURE 7. Dataset distribution per disponible.

Dataset. Used are the “Pothole Detection Dataset” [90] and the “Annotated Potholes Image Dataset [91], each of which

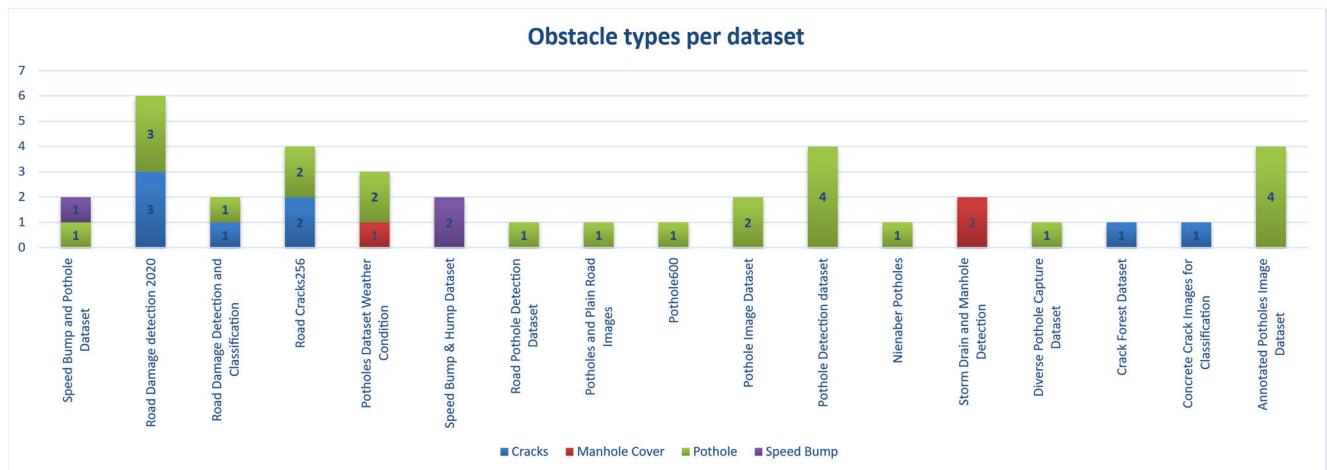


FIGURE 8. Obstacle types in each dataset.

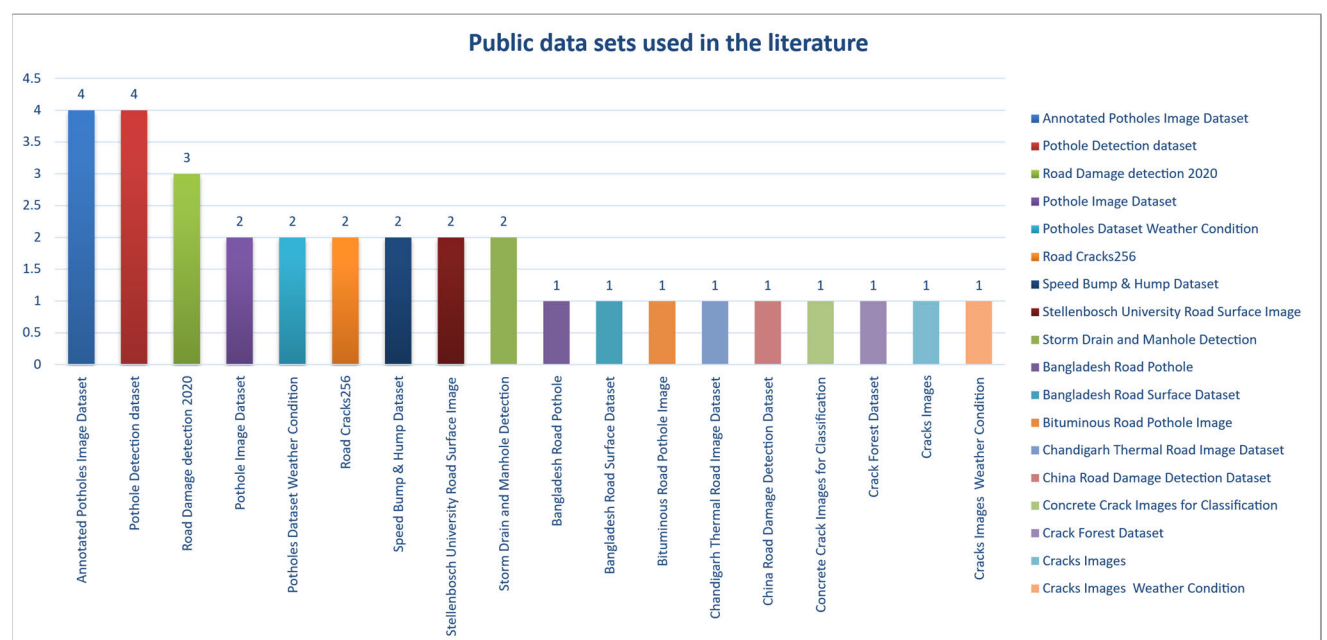


FIGURE 9. Public data sets used in the literature.

is used four times. Next, the Road Damage Detection 2020 [92] is used three times. Several datasets, such as the “Speed Bump & Hump Dataset” [93], the ‘Road Cracks256’ [51], the ‘Potholes Dataset Weather Condition’ [86] And others are each used twice. The graph also shows a diversity of less frequently used datasets, each appearing only once in the literature, such as “Road Damage Detection and Classification” [96], and others. This Fig. illustrates the range of datasets used by researchers to study road defects, highlighting the most popular datasets and the variety of data sources exploited. To deepen our analysis, let’s take a look at another chart that presents the datasets used according to the objective of the detection, classification, or both. The following illustrates this distribution.

This Fig. 10 shows the use of public datasets according to the purpose of the study, divided into three categories: Detection, Classification, and both aspects at once. For studies focused on. For detection, the “Annotated Potholes Image Dataset” [91] leads with three occurrences, followed by the “Potholes Dataset Weather Condition [86] Used twice. Other datasets used once include “Storm-drain and manhole detection.” [38], “Road Pothole Detection Dataset” [95], “Pothole Image Dataset” [94], and others. As far as classification studies are concerned, the “Pothole Detection Dataset” [90] is the most widely used, with three occurrences. Other datasets used once each include “Speed Bump and Pothole Dataset.” [45], “Speed Bump & Hump Dataset” [93], “Road Cracks256” [51], and “Pothole600” [98]. In addition,

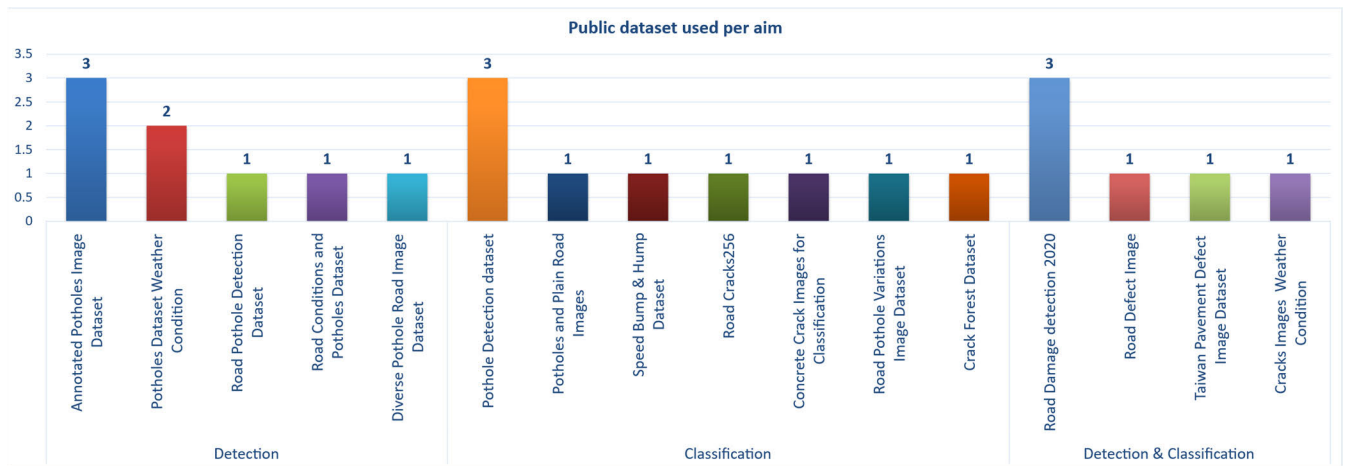


FIGURE 10. Public data sets were used according to the study objective.

TABLE 3. Public datasets used in the literature.

Dataset	REFS	TAILLE	CLASSE
Pothole Detection Dataset (PDD)	[87]	681	NR, P
Annotated Potholes Image Dataset (API-D)	[88]	665	P
Road Damage Detection 2020 (RDD-2020)	[89]	26 336	C, P
Speed Bump/Hump Dataset (SBH-D)	[90]	550	B
Road Cracks256 (RC256)	[51]	6000	P, C, NR
Potholes Dataset Weather Condition (PDWC)	[87]	2099	P, M
Pothole Image Dataset (PDI)	[91]	618	P
Storm-drain and manhole detection (SDMD)	[38]	297	M, S
Speed Bump and Pothole Dataset (SBP)	[45]	714	P, B, NR
Road Pothole Detection Dataset (RPDD)	[92]	4188	P, NR
Road Damage Detection and Classification (RDDC)	[93]	9053	C, P
Potholes and Plain Road Images (PPRI)	[94]	740	P
Pothole600 (P600)	[95]	1800	P
Nienaber Potholes (NP)	[96]	8101	P, C, B, R
Diverse Pothole Capture Dataset (DPCD)	[31]	940	P
Crack Forest Dataset (CFD)	[97]	118	C
Concrete Crack Images for Classification (CCIC)	[98]	20 000	C

for studies combining detection and classification, the “Road Damage detection 2020” dataset [92] is the most frequently used, with three occurrences. Other datasets used once in this category include “Storm-drain and manhole detection.” [38], “Road Damage Detection and Classification” [96]. So, we can summarize that some datasets, such as “Pothole Detection Dataset” [90] and “Road Damage Detection 2020” [92], are versatile and widely used in various studies of classification, detachment, or both, while other datasets are

specifically used for a single type of purpose. To continue our analysis, let’s take a look at another graph showing the datasets used according to the type of obstacle dealt with in the studies, whether potholes, cracks, speed bumps, and other road obstacles. This analysis will enable us to understand how the various types of obstacles are studied and modeled, and to see if there are trends or preferences for certain types of data sets in current research into road obstacle detection research. The following Fig. shows this distribution in detail.

Fig. 11 shows the use of public datasets according to the type of obstacle dealt with in the studies, divided into four categories: cracks, manholes, potholes, and speed bumps. For potholes, the “Pothole Detection Dataset” [90] and the “Annotated Potholes Image Dataset” [91] are the most widely used, with four hits each. Each. “Road Damage Detection 2020” [92] is also popular with three hits, followed by ‘Road Cracks256’ [51] and ‘Potholes Dataset Weather Condition’ [86] With two hits each. For cracks, the “Road Damage Detection 2020” dataset [92] is the most used, with three hits, followed by “Road Cracks256” with two hits each. [51] Used twice. Other datasets, such as “Road Damage Detection and Classification” [96] and “Crack Forest Dataset” [101], are used once each. detection” [38] It is the most frequently used with two occurrences, while” Potholes Dataset Weather Condition” [86] It is used once. In addition, for speed bumps, the “Speed Bump & Hump Speed Bump & Hump Dataset [93] is the most used, with two occurrences, while the “Speed Bump and Pothole Dataset” [45] It is used once. To deepen this analysis and gain a more detailed perspective, we will examine a Stacked bar chart. This type of chart shows the distribution of obstacles in each data set, allowing us to visualize how each type is represented within the different sets. This approach will facilitate an understanding of the diversity of obstacles studied and their distribution within the analysis. The following Fig. shows this distribution.

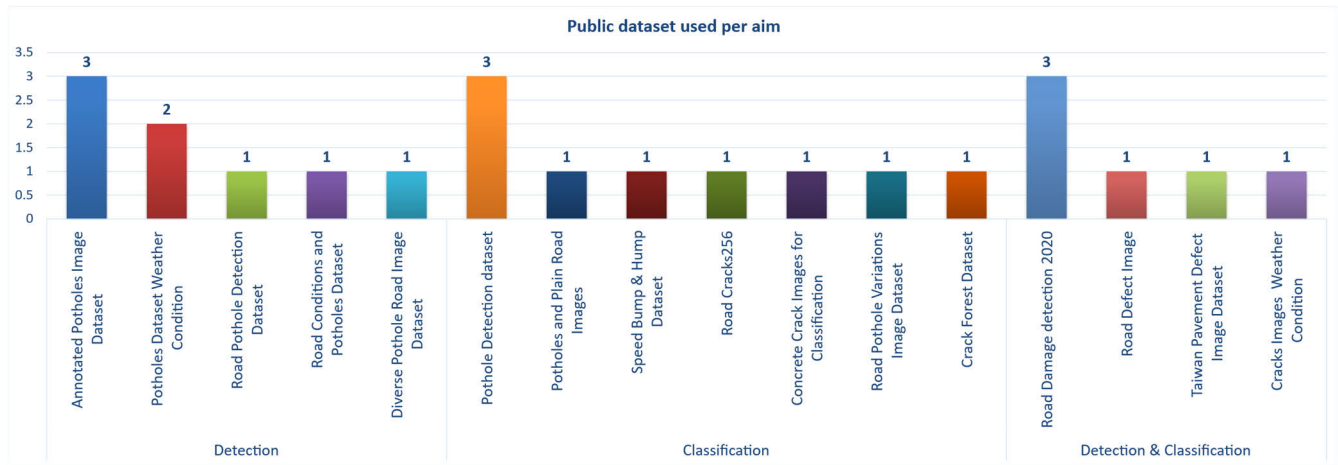


FIGURE 11. Data sets were used according to the type of obstacle dealt with in the studies.

For more clarity on the public dataset analysis, we provided Table 4 below, which shows the public datasets used in related works. Each entry includes the dataset’s name and a link to access it, typically hosted on platforms like Kaggle, Google Drive, or GitHub. The datasets are intended for use in research and machine learning applications focused on road damage detection, road maintenance, and improving infrastructure management.

2) DATA SOURCES USED

The diversity and quality of data sources are essential in road barrier research. The different data sources used in the literature offer a variety of perspectives and information that enrich our understanding of the subject. In this section, we will explore the data sources used by researchers to better understand their use and importance in studies. The following Fig. provides an overview of the main data sources used in the studies reviewed.

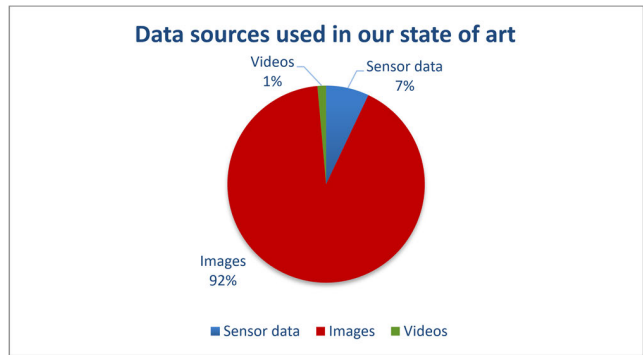


FIGURE 12. Data sources used in the studies reviewed.

Fig. 12 illustrates the distribution of data sources used in the literature for road obstacle detection. It shows a clear predominance of images, which account for 91.55% of the data sources used. This is due to the strong adoption of

TABLE 4. Public datasets are in the related work section.

Public dataset	LINK
Annotated Potholes Image Dataset	https://www.kaggle.com/datasets/chitholian/annotated-potholes-dataset
Annotated Potholes Image Dataset	https://www.kaggle.com/datasets/chitholian/annotated-potholes-dataset
Concrete Crack Images for Classification	https://www.kaggle.com/datasets/arnavr10880/concrete-crack-images-for-classification
Crack Forest Dataset	https://github.com/cuilimeng/CrackForest-dataset
Diverse Pothole Capture Dataset	https://makeml.app/datasets/potholes
Nienaber Potholes	https://www.kaggle.com/datasets/felipemuller5/nienaber-potholes-2-complex
Pothole Detection dataset	https://www.kaggle.com/datasets/atulyakumar98/pothole-detection-dataset
Pothole Image Dataset	https://www.kaggle.com/datasets/sachinpatel21/pothole-image-dataset
Pothole600	https://sites.google.com/view/pothole-600/dataset
Potholes and Plain Road Images	https://www.kaggle.com/datasets/virenbr11/pothole-and-plain-rode-images
Potholes Dataset	https://figshare.com/articles/figure/Potholes_dataset_with_YOLO_annotations/21214400/3
Weather Condition	https://drive.google.com/drive/folders/1wGmkq5aIXf1NNIK8f6Sow1S4AnZiLS9
Road Cracks256	https://data.mendeley.com/datasets/5ty2wb6gv/1
Road Damage Detection 2020	https://drive.google.com/file/d/1zu3yhcBt2WmQni5AEEsvj4UT8IOkSDOh/view
Road Damage Detection and Classification	https://www.kaggle.com/datasets/aminumusa/road-dataset
Road Pothole Detection Dataset	https://data.mendeley.com/datasets/xt5bjdhy5g/1
Speed Bump & Hump Dataset	https://drive.google.com/drive/folders/1ugaPdFtf5o2Wgreb2DT0BgVhdDk2uZAL
Speed Bump and Pothole Dataset	https://drive.google.com/drive/folders/1Eqltuuv0pPdGQhTt_u96xnR43Gw66q2M
Storm Drain and Manhole Detection	

computer vision techniques, which rely heavily on image analysis to detect and classify road obstacles such as potholes,

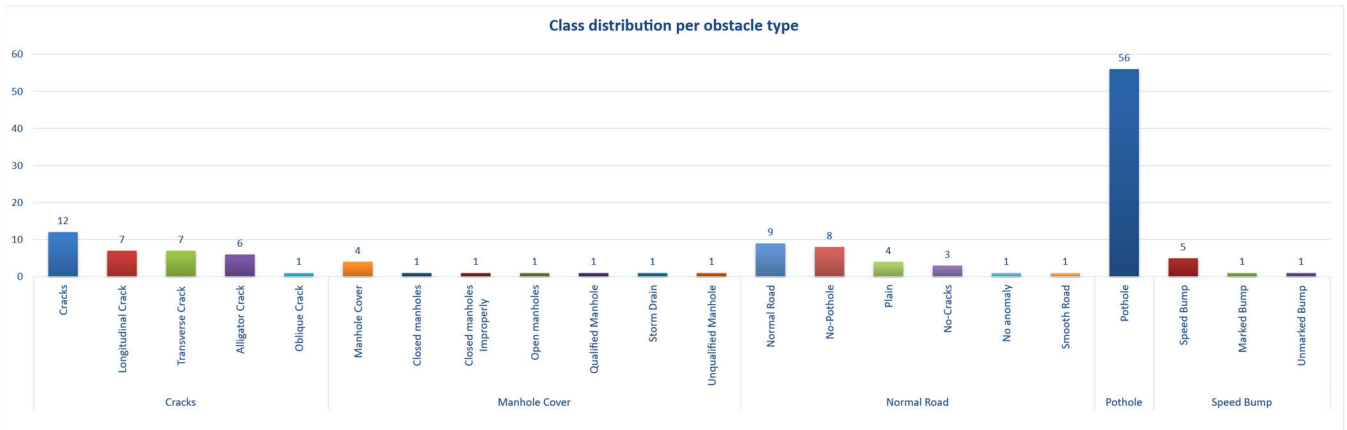


FIGURE 13. Classes of obstacles treated.

cracks, speed bumps, and manholes. Sensor data then makes up 7.04% of data sources. These data typically include accelerometer readings, vibration measurements, and other onboard sensors. They are often used alongside images to improve the accuracy and robustness of detection systems. Detection systems. Finally, videos represent only a small portion of 1.41% of data sources. Data sources. Although videos can provide rich, contextual information for real-time obstacle detection in real-time, they require greater processing resources and an adequate infrastructure for their analysis. Analysis of this Fig. reveals that images are by far the preferred data source for road obstacle detection, probably due to their ease of acquisition and the maturity acquisition and the maturity of image processing algorithms. Sensor data, although less common, plays an important role in applications where more precise, contextual detection is required. Video, although underutilized, represents untapped potential for future applications.

3) OBSTACLE CLASSES

Roads can present a variety of obstacles, from cracks and potholes to manholes and speed bumps. Analyzing the diversity and frequency of these classes helps to focus research efforts. The aim is to provide an overview of the class diversity of each type of obstacle, which is of interest to the understanding and development of road defect detection and classification systems. The following Fig. shows the class distribution of the datasets used in the literature, according to the different types of obstacles dealt with.

Fig. 13 shows that for potholes, the “Pothole” class largely dominates with 56 occurrences, underlining the importance of this category in the literature. Normal roads are then categorized into “Normal Road” (9 occurrences), “No-Pothole” (8 occurrences), “Plain” (4 occurrences), “No-Cracks” (3 occurrences), “Smooth Road”, and “No anomaly”, each with 1 occurrence. As far as cracks are concerned, the generic class “Cracks” predominates with 12 occurrences, followed by “Transverse Crack” and “Longitudinal Crack” with

7 occurrences each. Alligator Crack” and “Oblique Crack” are also mentioned, with 6 and 1 occurrences, respectively. Manholes are mainly represented by the “Manhole Cover” class, with 4 occurrences. Other less frequent classes include “Unqualified Manhole”, “Storm Drain”, and “Qualified Manhole”, as well as various mentions of open or closed covers, each appearing only once. Finally, for speed bumps, the “Speed Bump” class is mentioned 5 times, with “Unmarked Bump” and “Marked Bump” each mentioned once. Analysis of this graph highlights the diversity and frequency of the different classes of road obstacles studied in the literature. Cracks and potholes stand out as the most detailed types of obstacle obstacles, with a wide variety of sub-classes, particularly potholes, which are potholes, which are widely studied and documented. Normal roads and manholes also feature several sub-classes, although fewer in number. Speed bumps, although less frequent, show a clear classification.

D. PREPROCESSING TECHNIQUES APPLIED TO DATA SETS (RQ4)

In data analysis, data pre-processing plays an essential role in guaranteeing the quality and efficiency of models. Pre-processing techniques are essential for preparing data sets for use in analytical processes and learning algorithms. They enable data to be cleaned, normalized, augmented, and transformed raw data into a form more suitable for analysis, improving model performance and accuracy. This section explores pre-processing techniques [99] applied to data sets such as Data Augmentation, Annotation, Resizing, Data Filtering, Noise reduction, Cropping, Normalization, Data Conversion, Data Balancing, Resampling, Feature extraction, Segmentation, Image Denoising, Radiometric Correction, Image transformation, Annotation Correction, Multiband Fusion, Image Conversion, Data interpolation, Geometric correction. The following Fig. illustrates the pre-processing techniques used in the literature.

Fig. 14 shows two graphs providing a detailed perspective on the pre-processing techniques used in road obstacle

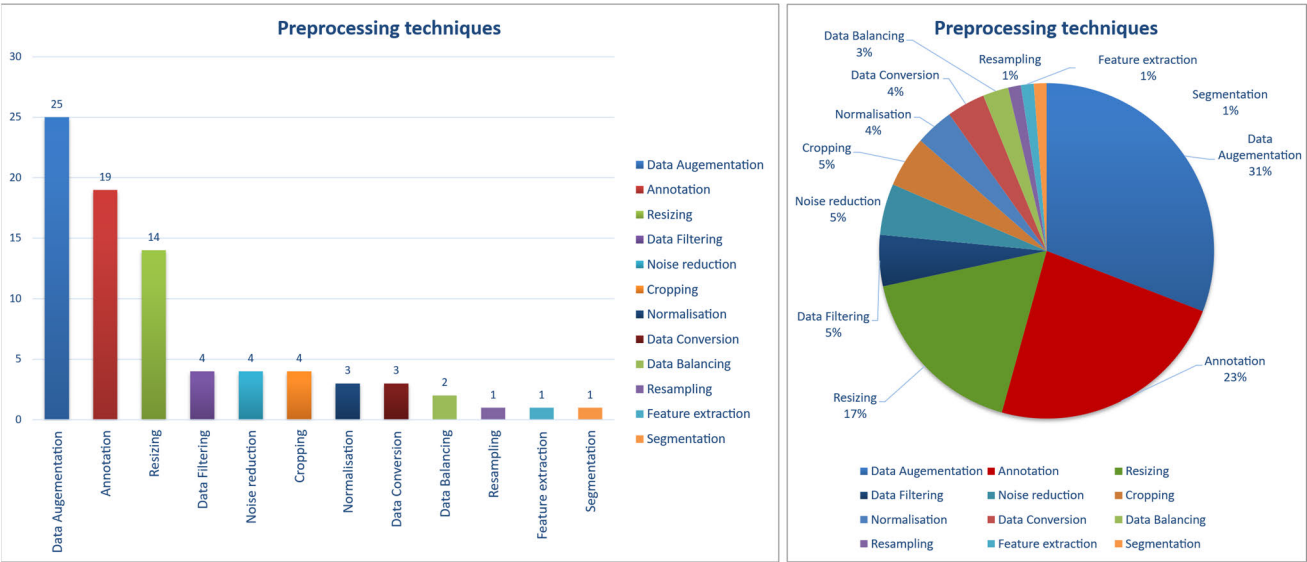


FIGURE 14. Pre-processing techniques used in the literature.

detection. The first graph, a bar chart, and the second, a pie chart, highlight the frequency and proportion of each method used. Data augmentation appears to be the most frequently employed technique, used by 25 articles (30.86%). This underlines the importance of this technique in improving the diversity and robustness of data sets, which is essential for the development of high-performance deep learning models. By enriching the available data with various transformations (such as rotations, translations, and changes in brightness), data augmentation helps prevent overlearning and improve model generalization. Annotation, used in 20 articles (24.69%), is also an important technique. It involves labeling raw data to create datasets that can be used for object detection. The high frequency of this technique reflects the need for well-labeled data to enable models to correctly detect and classify road obstacles. Then, resizing, applied in 14 articles (17.28%), is another key technique. It is essential for normalizing image dimensions, ensuring their compatibility with CNN architectures. This standardization is essential for efficient image processing by deep learning algorithms. Other techniques, such as noise reduction, cropping, and data filtering, each applied in four papers (4.94%), also play important roles. Cropping improves data quality by removing irrelevant parts of the images, while noise reduction and data filtering eliminate superfluous or harmful information. Normalization, used in three articles (3.70%), adjusts pixel values to a uniform scale, improving data consistency. Finally, less common techniques such as class balancing, image transformation, and segmentation, each used in one or two articles, show that these methods are applied in a more specific and targeted way, depending on the precise needs of the analysis and the quality of the data. This Fig. highlights the importance of the various pre-processing techniques in road obstacle detection. Data augmentation, annotation, and

resizing are the dominant techniques, reflecting their essential role in improving data quality and quantity for training deep learning models. The other techniques, although less frequently used, also contribute also make significant contribution to optimizing data for more accurate and robust analysis. So, given that data augmentation is the most widely used technique, we're going to take a closer look at the different methods applied for this technique. The following Fig. presents the data augmentation techniques used in the selected studies.

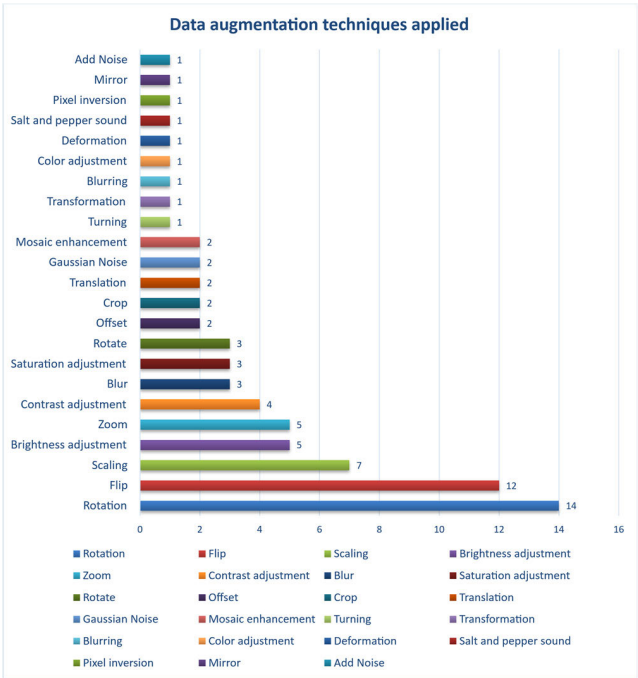


FIGURE 15. Data augmentation techniques are used in studies.

Fig. 15 shows the distribution of data augmentation techniques used in several studies to improve the performance of road obstacle detection models. Each technique is represented by the total number of papers in which it was applied. Image rotation is the most frequently used technique, cited in 17 articles. This popularity is explained by the ability of rotation to diversify the viewing angles of training images, helping models recognize obstacles from different perspectives. Flipping is the second most common technique, mentioned in 13 articles. It enables mirror images to be created, thus increasing the diversity without introducing major distortions. This helps models to generalize better on new and varied data. After scaling, applied in 7 articles, is also an essential technique. It makes it possible to adjust the size of objects in images, helping models recognize obstacles at different distances. Zoom and brightness adjustment techniques are each used in 5 items. Zooming helps simulate variations in distance while adjusting brightness adjustment makes models robust to varying lighting conditions. The blurring and contrast adjustment, cited in 4 articles each, contributes to the robust models by simulating low-quality image conditions or variations in contrast, respectively. Similarly, saturation adjustment, which helps to manage color variations in images, was applied in 3 papers. Other techniques such as translation, cropping, offset, Gaussian noise, and mosaic enhancement, each mentioned in 2 articles, are used less frequently but still play a crucial role in the diversification of training data. Finally, techniques such as transformation, mirroring, pixel inversion, distortion, salt-and-pepper noise, and color adjustment, each applied in a single article, show more specialized and less common approaches. This Fig. highlights the variety of data augmentation techniques used to enrich training datasets. Rotation, flipping and scaling are the most common techniques, reflecting their effectiveness in improving data diversity and robustness of the data. Having reviewed the different pre-processing techniques applied in the literature, we will now focus on pre-processing techniques specific to each public dataset used by researchers. This approach will enable us to better understand how each dataset has been prepared and the specific methods used to optimize data quality before analysis.

Fig. 16 illustrates the pre-processing techniques applied to the public data sets used, with each color representing a specific technique. For example, for the “Pothole Detection dataset” the techniques applied include data augmentation (used in 3 articles), class balancing (1 article), and resizing (2 articles). For the “Annotated Potholes Image Dataset” three papers used data augmentation, while data conversion, resizing, and cropping were each applied once. The “Road Damage detection 2020” dataset was analyzed in three papers, with data augmentation, data filtering, and normalization techniques each used once. For the other sets, the most frequently used technique was data augmentation

E. TECHNIQUES AND MODELS USED IN SELECTED ARTICLES (RQ5)

The methodological approaches and analytical tools chosen by researchers directly influence the quality and relevance of the results obtained. This section explores the different techniques and models used in the selected articles, highlighting the strategies adopted to address the specific challenges associated with the study of road obstacles. Fig. 17 presents graphs that provide an analysis of the techniques used in the studies reviewed, with a focus on three main categories: Deep Learning, Machine Learning, and a combination of the two. The interpretation of the results, trends, and preferences in the field is examined. The first graph, a bar chart, clearly shows that Deep Learning is the predominant technique, with 61 studies using it exclusively. This overwhelming majority indicates a strong preference for Deep Learning, probably due to its advanced ability to handle complex tasks such as object detection and classification. Of objects. The high number of articles devoted to this technique also reflects the maturity and effectiveness of Deep Learning methods in the field studied. On the other hand, only 7 studies combine Deep Learning and Machine Learning. This suggests that, although the combination of the two approaches may offer advantages, it is less common in current research. This could be due to the complexity of integrating two different methodologies, or perhaps the fact that the performance gains do not always justify the extra effort required. Finally, Machine Learning alone was used in only 2 studies. This Fig. underlines the fact that, in the context of the studies examined, traditional Machine Learning is less often considered as an effective solution compared to Deep Learning. This trend can be explained by the limitations of traditional Machine Learning techniques when it comes to handling tasks requiring deeper data analysis. The second chart, a pie chart, offers another perspective by presenting the relative proportions of the techniques used. Deep Learning accounts for 87.14% of studies, confirming its dominance. The combination of Deep Learning and Machine Learning accounts for 10%, while Machine Learning alone represents only 2.86%. These percentages reinforce the idea that Deep Learning is widely preferred, while combined approaches and Machine Learning alone remain minority choices. These graphs reveal that Deep Learning is largely dominant in the studies examined, reflecting its perceived superiority in complex detection and classification tasks. The combination of Deep Learning with Machine Learning is used to a limited and Machine Learning alone is even less common. These trends underline the growing importance of Deep Learning in the field under study, and the ability to explore and optimize its applications.

Having examined the techniques adopted, it is important to look at the various models used in the literature to study road obstacles. These models include statistical approaches, machine learning algorithms, and image processing methods. The following Fig. provides an overview of the different models used in the studies reviewed.

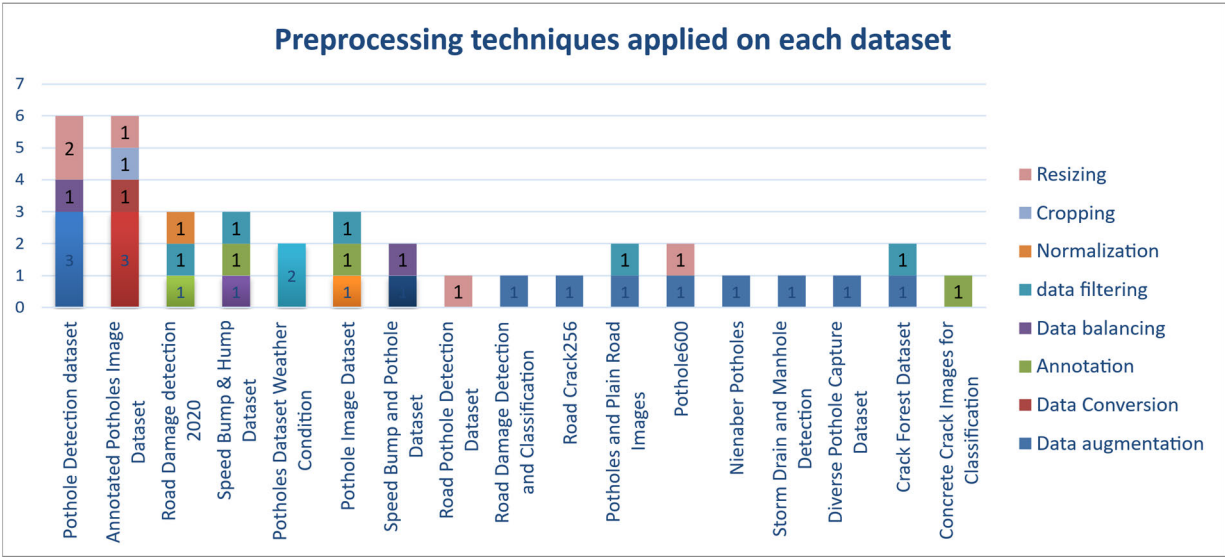


FIGURE 16. Pre-processing techniques applied to each dataset.

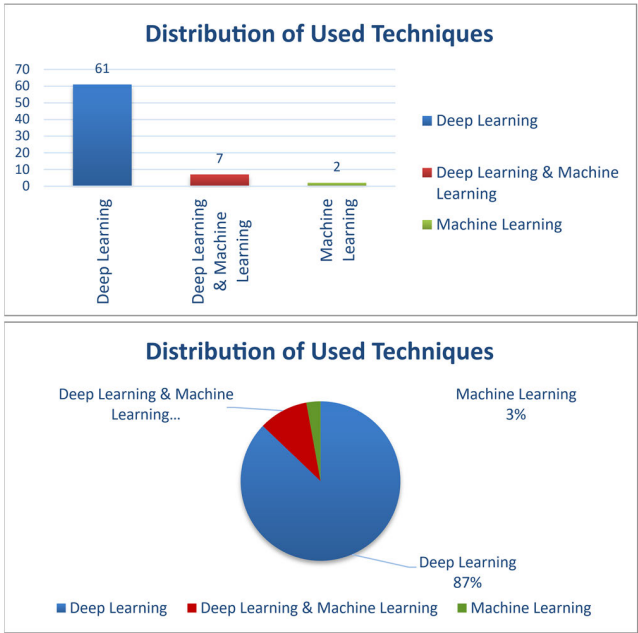


FIGURE 17. Techniques used in the studies reviewed.

Fig. 18 shows that the YOLO (You Only Look Once) model is clearly the most.

Popular, with 29 occurrences in the studies reviewed. This dominance can be attributed to the ability of YOLO to perform real-time object detection, an important feature for many practical applications. YOLO is known for its speed and efficiency, which explains why it is widely adopted. CNN is the second most important in 20 studies. CNNs are widely recognized for their outstanding performance in image classification and object detection tasks. Their flexibility and

ability to learn from hierarchical representations of visual data make them a preferred choice. Faster R-CNN, ResNet, and SSD (Single Shot MultiBox Detector) follow with 10, 9, and 8 occurrences, respectively. Faster R-CNN is appreciated for its accuracy in object detection, although it is slower than YOLO. ResNet (Residual Networks) is a deep image classification model that can overcome the problem of degradation in very deep networks. SSD, meanwhile, offers a good compromise between speed and accuracy, making it popular in applications where processing time is critical. Other models, such as VGG, Inception, and MobileNet, show moderate use with 6, 6, and 4 occurrences, respectively. VGG is known for its simplicity and depth, Inception for its innovative architecture to capture multi-scale features, and MobileNet for its efficiency in mobile and embedded applications, thanks to its low resource consumption. The presence of models such as SVM (Support Vector Machine), Sparse R-CNN, and DenseNet, with 3 and 2 occurrences, shows a diversity of approaches in studies, although their use is less frequent compared to leaders like YOLO and CNN. Finally, a multitude of models appear only once in the studies. This includes approaches such as LSTM (Long Short-Term Memory) networks, traditional machine learning algorithms like Naive Bayes and KNN (k-Nearest Neighbors), as well as specialized methods such as bilateral filters and ensemble learning techniques. This long queue of models used only once illustrates the diversity of approaches explored by researchers, often adapted to specific use cases or particular constraints.

The Fig. reveals a clear predominance of Deep Learning models, particularly YOLO and CNN, in the studies examined. These models are preferred for their ability to efficiently handle object detection and classification tasks. However, the diversity of models used shows that researchers continue to

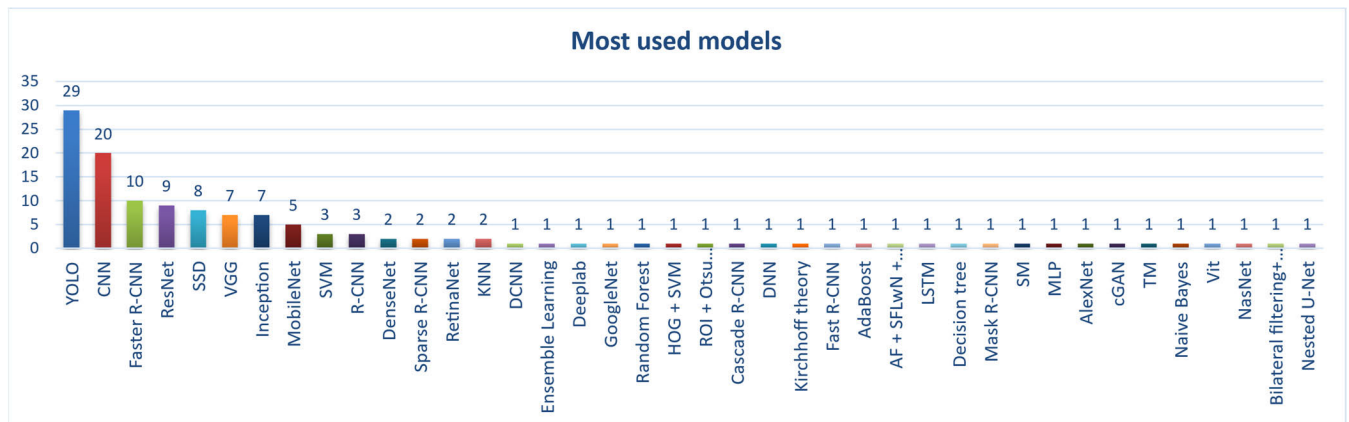


FIGURE 18. Models used in the studies reviewed.

explore and adapt a variety of techniques to meet the specific needs of their applications. This exploration contributes to the continuous evolution and improvement of performance in the field of object detection and classification. After examining the different models used in the literature, we will now analyze the distribution of these models into three distinct categories: deep learning (DL) models, machine learning (ML) models, and those combining the two approaches (DL/ML). The following Fig. shows the distribution of models used by researchers according to technique.

Fig. 19 shows an analysis of the models used in the studies, classified according to three techniques: deep learning, machine learning, and a combination of both. Deep learning is the dominant model, with YOLO (You Only Look Once) at the top of the list, used in 28 studies. This model is particularly popular for object detection and recognition tasks because of its speed and accuracy. CNN comes in second place with 17 uses, followed by other models like Faster R-CNN, SSD, and ResNet, each being used in several studies. These models are valued for their ability to extract complex data characteristics, particularly in the field of image analysis. Machine learning models are less represented than deep learning models. The SVM (Support Vector Machine), KNN (K-Nearest Neighbors), and Random Forest are mentioned, each appearing in a single study. These models are often used for supervised learning tasks, such as classification or regression, and are valued for their robustness and simplicity. Finally, some studies use a combination of deep learning and machine learning techniques. This may indicate a tendency to integrate different approaches to leverage the strengths of each method. For example, CNN and YOLO are used in combination with machine learning techniques in some studies, suggesting an exploration of synergies between approaches for specific tasks.

Fig. 20 shows the different versions of the models used in the studies reviewed. The YOLO (You Only Look Once) model family is by far the most represented, with a total of 45 articles. Among these versions, YOLOv3 stands out

with 15 uses, followed by YOLOv5 and YOLOv4, used in 11 and 7 studies, respectively. Other versions of YOLO, such as YOLOv8 and YOLOv7, YOLOv6 are much less common, each being mentioned in two or only one article. For other models, ResNet is the second most widely used group. The ResNet50 version dominates this category, appearing in 6 studies. The other versions of ResNet, such as ResNet50V2, ResNet18, ResNet152V2, and ResNet101, are each used in a single study. The VGG (Visual Geometry Group) family is also well represented, with VGG19 being the most popular, used in 4 studies, followed by VGG16 and VGG11, present in two and one article, respectively. For the Inception family, the most widely used version is InceptionV3, appearing in 6 studies, while InceptionV2 is mentioned in two articles. MobileNetV2 is used in two studies, and DenseNet presents the versions DenseNet121 and DenseNet201, each appearing in one study. The YOLO model, especially the YOLOv3, YOLOv5, and YOLOv4 versions, is widely preferred by researchers. ResNet, VGG, and Inception are also popular, but to a lesser extent, with a marked preference for ResNet50, VGG19, and InceptionV3, respectively. MobileNet and DenseNet are the least represented among the models studied. These trends indicate a predominance of YOLO models in the current literature, probably due to their performance and efficiency in object detection tasks.

F. THE PERFORMANCE METRICS ADOPTED IN THE LITERATURE (RQ6)

The evaluation of the performance of the models ensures their effectiveness and reliability. Researchers use various performance measures to analyze and compare the results of their models. These measures allow for the quantification of precision, sensitivity, specificity, and other relevant aspects of their approaches. Understanding the different metrics used in the literature helps us to better interpret results and identify the best practices in this area. The following Fig. provides an

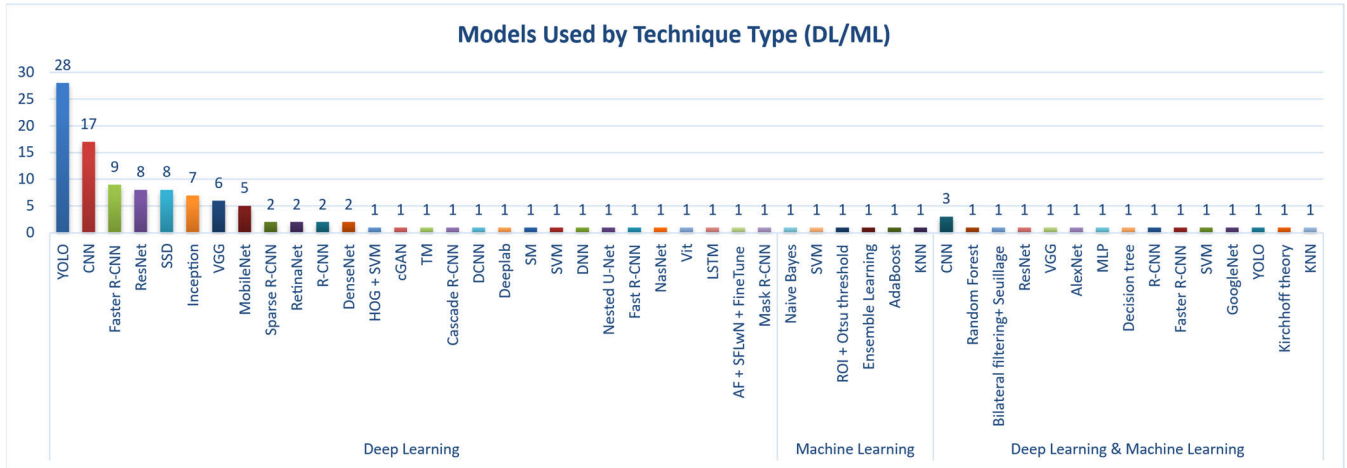


FIGURE 19. Models used in studies examined by technique (DL/ML).

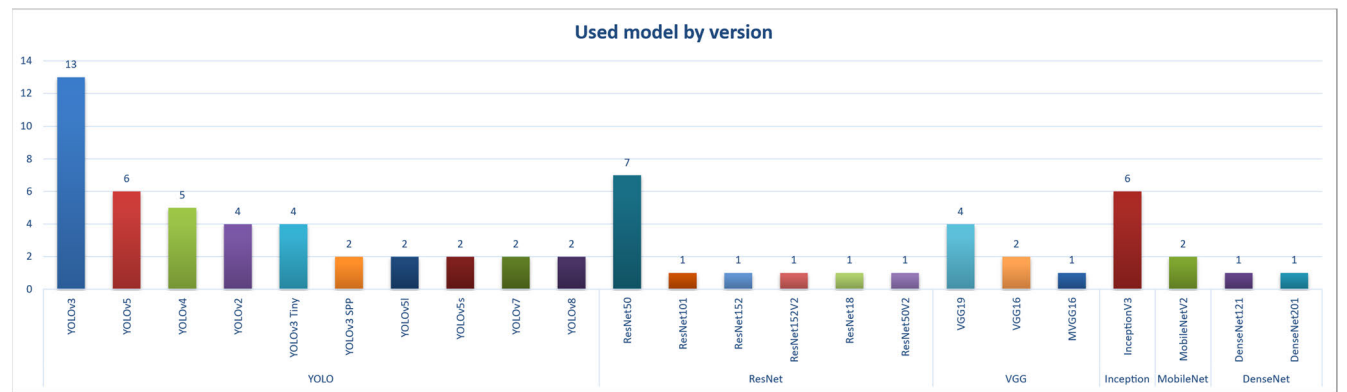


FIGURE 20. Model versions used in the studies reviewed.

overview of the key performance measures used in the studies reviewed.

Fig. 21 shows two graphs that illustrate the performance measures used in the literature to evaluate obstacle detection and classification models for automated driving. The graphs clearly show that precision is the most frequently used metric, appearing in 36 articles (23.84%), closely followed by accuracy (Accuracy), also present in 36 articles (23.84%). Recall is used in 30 articles (19.87%), while the average precision (mAP) and F1-score are each present in 22 articles (14.57%). The Intersection Over Union (IoU) is the least used metric, with only 5 occurrences (3.31%). Analysis of these graphs shows a clear trend in the literature to focus on accuracy and precision as key performance measures for obstacle detection and classification models. These metrics are important for assessing the ability of models to correctly identify barriers and properly rank predictions. Recall, also widely used, is essential to assess the model's ability to detect all relevant barriers. Metrics like mAP and F1-score, although important, are less commonly used, probably due to their relative complexity or relevance for specific aspects of the

models. Finally, the IoU, often used in more specialized or advanced contexts, remains the least popular metric. This Fig. illustrates the preferences and priorities of researchers in evaluating obstacle detection models, highlighting the importance of accuracy and precision in developing reliable and efficient automated driving systems.

For further analysis, it is essential to consider performance measures about the specific objective of each study. Some research focuses on the detection of barriers, others on their classification, while some combine both approaches. By examining the performance measures used for these objectives, we can better understand the evaluation criteria applied by researchers. To make it clearer and understandable, it is crucial to showcase the equation used to calculate these performance metrics:

- Precision: mainly used to assess the quality of correct predictions achieved by the model, and it is calculated using this formula:

$$Pr = \frac{\text{classified actual positive}}{\text{all instances classified as positive}} = \frac{TP}{TP + FP}$$

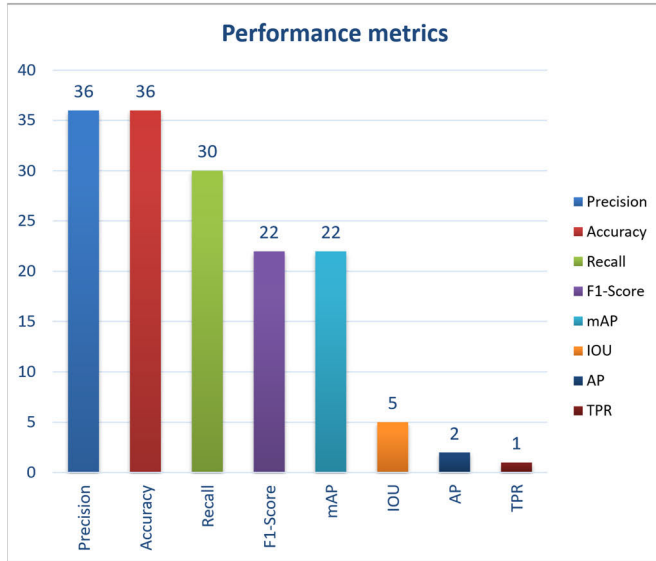


FIGURE 21. Performance measures used in literature.

where FP: false positive, TP: true positive, FP: false positive, FN: false negative.

- Recall: This metric answers this question: “What is the probability of the real positive that was predicted as positive?” It can be calculated using the formula below:

$$Re = \frac{\text{classified actual positive}}{\text{all actual positive}} = \frac{TP}{TP + FN}$$

- F1 score: This metric combines precision and recall into a single score to show how well the model identifies positive cases. It gives a balanced view of both and is calculated using the following formula:

$$F1_score = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2 \times TP}{2 \times TP + FP + FN}$$

- IoU: is a metric used to measure how well the predicted boxes match the actual ones in object detection. It shows how much the predicted and actual boxes overlap, helping to distinguish correct detections from incorrect ones. If P is the area of the predicted box and A is the area of the actual box, IoU is calculated as:

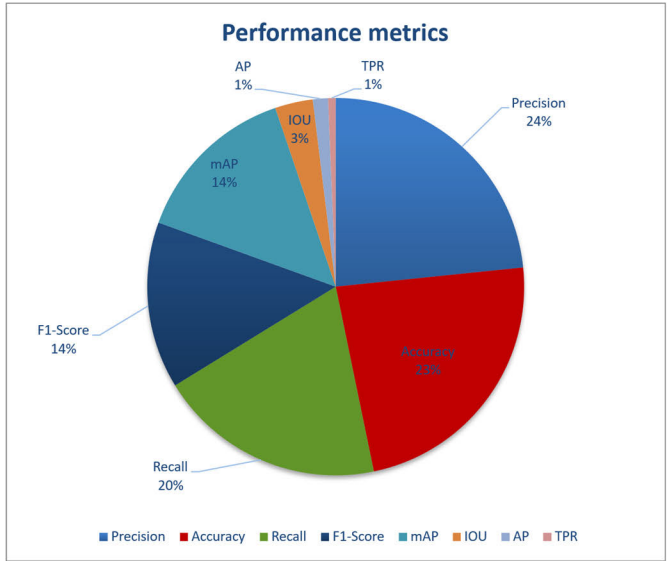
$$IoU = \frac{P \cap A}{P \cup A}$$

- Mean average precision (mAP): It shows how well the model detects objects by averaging the precision at different IoU levels. It's calculated from the area under the precision-recall curve, calculated using this formula:

$$mAP = \frac{1}{n} \sum_{k=1}^n (AP_k)$$

where AP_k : classAvgPr. And n is the class count.

Accuracy: It answers the question: “How frequently does the model make correct predictions?”. It can be calculated



using this formula:

$$Acc = \frac{\text{correct predictions}}{\text{total predictions}} = \frac{TP + TN}{TP + TN + FP + FN}$$

The following Fig. provides an overview of the performance measures used in the studies, classified by their objectives.

The Fig. 22 shows two separate graphs, each providing information on performance measures used in different studies. For the bar graph, it is observed that for detection studies, precision is most used (19 times), followed by mAP (14 times), recall (13 times) and accuracy (11 times). For classification studies, accuracy is the most frequently used measure (20 times), followed by recall (10 times) and precision and F1-score (each 9 times). In studies combining detection and classification, the F1-score is most used (9 times), followed by precision (8 times), mAP (8 times), recall (7 times), and accuracy (5 times). The second graph, consisting of three pie charts, shows the percentage distribution of performance measures used according to the study objective. In detection studies, the accuracy is 29.23%, followed by mAP (21.53%), recall (20.00%) and accuracy (16.92%). For classification studies, the accuracy dominates with 41.66%, followed by recall (20.33%) and then F1-score and precision (18.75% each). Finally, for studies combining detection and classification, the measurements are more varied with F1-score (23.68%), accuracy (21.05%), mAP (21.05%), and recall (18.42%). Analysis of the graphs reveals that for detection studies, accuracy and mAP are the predominant measurements, indicating the importance of assessing the quality of positive predictions and averaging accuracies across different classes. In addition, accuracy is the most frequently used performance measure for classification studies, which is consistent to measure the proportion of correct predictions. In studies combining detection and classification,

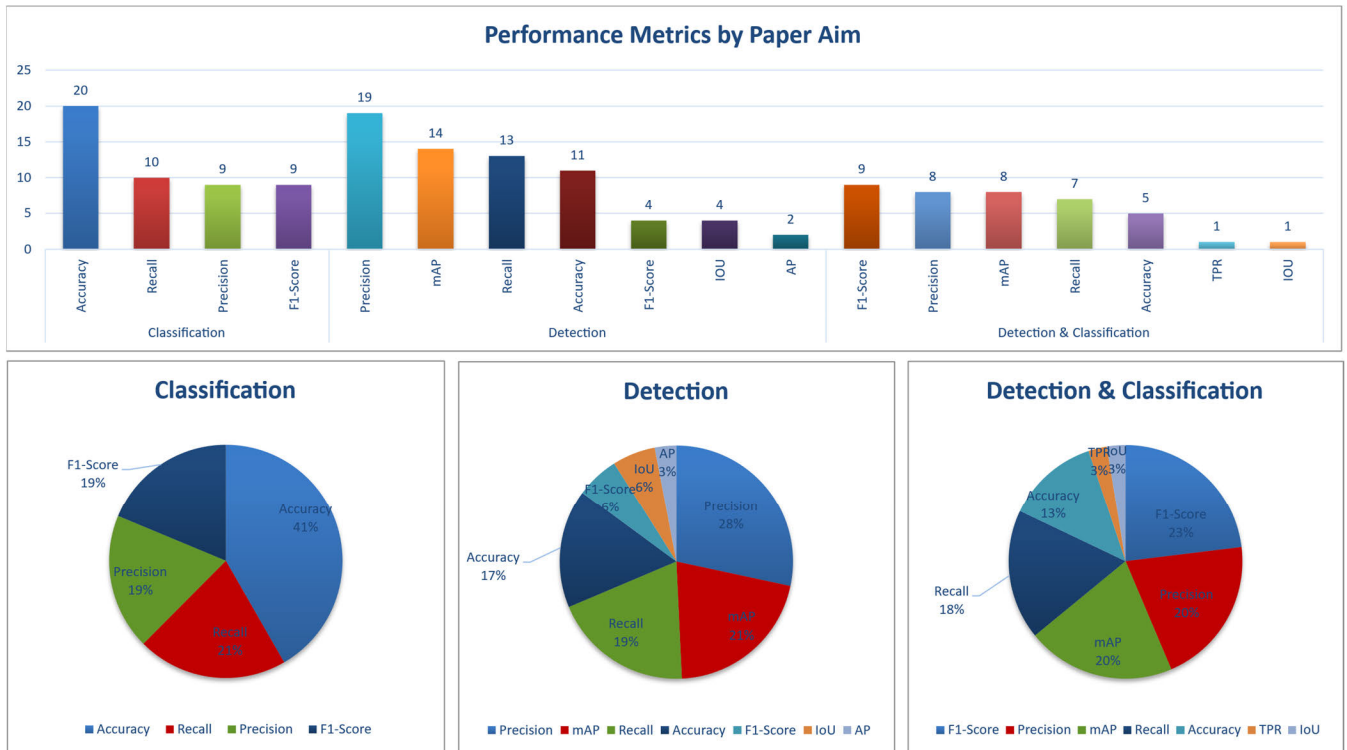


FIGURE 22. Performance measures used according to the study objective.

the F1-score is preferred, highlighting the need for a balance between accuracy and recall. The choice of performance measures varies according to the specific objective of the study, with each measure offering different perspectives on the quality of machine learning models. These variations reflect distinct priorities in performance evaluation, whether for classification, detection or a combination of both.

G. THE RESULTS OF THE MODELS USED IN THE ARTICLES STUDIED (QR7)

1) THE RESULTS OF MODELS USED FOR THE DETECTION OF ROAD OBSTACLES

The models developed for road obstacle detection are evaluated on their ability to accurately and reliably identify the different types of obstacles present on roads. In this section, we will analyze the results obtained by these models, to better understand the effectiveness of the different approaches used by researchers and to identify the most promising models in this field. The following table 5 gives a detailed summary of the results obtained for road obstacle detection

The above Table 5 summarizes the performance of the different detection models for Road barriers using multiple data sets. To visualize these results more intuitively, we will then present two box plots (box plot). A box plot is a type of chart that shows the distribution of a data set by highlighting the median, quartiles, and outliers. It allows for a visual comparison of the performance of different models in terms

of variation and central trend. We will examine two separate graphs that present the most used performance measures for road obstacle detection: precision (Precision) and mean accuracy (mAP). The following figures show the accuracy and mAP values obtained by different models applied in the literature:

Analysis of the performance results of different models of road obstacle detection reveals significant variations in precision and mAP. Emphasis is placed on these two metrics to evaluate the best performance of applied models. Based on Table 5 and Fig. 23 and Fig.24, the best results are from the YOLOv8 model applied to the “Annotated Potholes Image Dataset,” which has an accuracy of 100% and a mAP of 99.50%. These figures indicate that the model is extremely effective at detecting potholes with very high accuracy and medium-high precision. In addition, the YOLOv5s model also shows impressive performance on the same dataset, with 100% accuracy and a mAP of 84.70%. This result is slightly lower than that of YOLOv8, but still very good. Another notable model is the SSD+MobileNetV1, applied to the <<Speed Bump Image Dataset, with an accuracy of 95.67%. Another notable result is the YOLOv5 model, applied to another private dataset called “Road Conditions and Potholes Dataset”, which has a mAP of 99.90%. This exceptional mAP demonstrates the model’s ability to detect potholes with near-perfect average accuracy in all classes of objects. The YOLOv4 model used on the <<Diverse Pothole Road Image Dataset>> also has a mAP of 93.30%, demonstrating

TABLE 5. Road obstacle detection results.

Refs	Dataset	Model	Precision	mAP	Rappel	Accuracy	IOU	F1-SCORE
[17]	Pothole Images under Various Scenarios	YOLOv3	90,00%	65,05%	41,00%	-	-	-
[18]	Google Pothole Diversity	ROI + Otsu Threshold	-	-	-	82,00%	-	-
[19]	Annotated Potholes Image Dataset	YOLOv8	100%	99,50%	100%	100%	-	100%
		YOLOv5s	100%	84,70%	90,00%	95,00%	-	81,00%
		YOLOv5l	100%	85,30%	86,00%	98,00%	-	81,00%
[20]	Track Training Data Repository	CNN	-	-	-	88,60%	-	-
[21]	Pothole and Normal Road Smartphone Image	YOLOv2	87,00%	-	89,00%	-	-	-
		Faster RCNN(VGG16)	79,00%	-	80,00%	-	-	-
		Faster RCNN(ResNet18)	81,00%	-	83,00%	-	-	-
[22]	Road Vibration and Pothole Data Set	CNN	-	-	-	93,24%	-	-
[23]	Speed Bump Image Dataset	SSD+MobileNetV1	95,67%	-	97,14%	95,63%	-	96,40%
[24]	Pothole Detection dataset	CNN	-	-	-	94,00%	-	-
		InceptionV3	-	-	-	92,00%	-	-
[25]	Road Pothole Detection Dataset	TM + SM	-	-	-	-	-	-
[26]	South African Pothole Image Dataset	Faster RCNN	7,70%	41,50%	66,00%	-	-	13,70%
		SSD	4,43%	18,50%	32,60%	-	-	7,60%
		YOLOv3	34,70%	34,70%	32,00%	-	-	42,00%
[27]	Pavement View Pothole Image Dataset	YOLOv3	92,00%	83,43%	75,00%	64,45%	-	-
		YOLOv3 Tiny	94,00%	79,33%	64,00%	53,26%	-	-
		YOLOv3 SPP	94,00%	88,93%	82,00%	72,10%	-	-
[86]	Potholes Dataset Weather Condition	YOLOv3	63,72%	49,08%	48,52%	49,08%	-	-
[28]	Diverse Pothole Road Image Dataset	YOLOv3	-	88,90%	-	-	63,50%	-
		YOLOv4	-	93,30%	-	-	74,10%	-
[29]	Surface Roughness and Speed Signal	CNN	-	-	-	96,38%	-	-
		Kirchhoff theory	-	-	-	89,73%	-	-
		KNN	-	-	-	82,74%	-	-
[30]	Annotated Potholes Image Dataset	YOLOX-nano	-	85,60%	-	-	-	-
[31]	Diverse Pothole Capture Dataset	YOLOv5Yl	86,43%	63,43%	-	-	-	-
	+	YOLOv5Ym	86,96%	61,54%	-	-	-	-
	Annotated Potholes Image Dataset	YOLOv5Ys	76,73%	58,90%	-	-	-	-
		Faster RCNN	91,90%	64,12%	-	-	-	-
		VGG16	69,80%	35,30%	-	-	-	-
		MVGG16	81,40%	45,40%	-	-	-	-
		MobileNetV2	63,10%	30,50%	-	-	-	-
		InceptionV3	72,30%	32,30%	-	-	-	-
[84]	Multivariate Pavement Dataset	CNN	-	-	-	95,00%	-	-
[44]	Road Pothole Image	SSD	73,00%	32,50%	37,50%	-	-	-
		YOLOv3	84,00%	83,62%	78,00%	-	-	-
		YOLOv4	85,00%	85,39%	81,00%	-	-	-
[32]	Two-Wheeled Vehicle Pothole Images + Vehicle Sensor DataSet	DNN	-	-	-	96,70%	-	-
		SVM	-	-	-	92,90%	-	-
[33]	Stellenbosch University Road Surface Image	Faster RCNN	-	-	-	-	-	-
		InceptionV3	-	-	-	-	-	-
[34]	Potholes Dataset Weather Condition	RCNN	48,70%	44,60%	-	-	-	-
		Fast RCNN	50,80%	46,80%	-	-	-	-
		Faster RCNN	52,30%	48,40%	-	-	-	-
		Mask RCNN	52,10%	48,90%	-	-	-	-
		Cascade RCNN	54,20%	50,50%	-	-	-	-
		Sparse RCNN	55,40%	51,40%	-	-	-	-
		YOLOv3	63,70%	49,10%	-	-	-	-
		YOLOv4	65,10%	50,10%	-	-	-	-
		YOLOv4-csp	66,40%	51,20%	-	-	-	-

TABLE 5. (Continued.) Road obstacle detection results.

		YOLOv5m	68,80%	53,00%	-			
		YOLOv5l	70,60%	54,30%	-			
		YOLOv6m	72,10%	54,80%	-			
		YOLOv6l	72,40%	55,00%	-			
		YOLOv7	73,20%	55,60%	-			
[35]	Port Harcourt Pothole Image	YOLOv3	97,00%	-	-	-	-	-
[36]	Road Conditions and Potholes Dataset	YOLOv5	-	99,90%	-	-	-	-
[37]	Speed Bump & Hump Dataset	cGAN	-	-	-	-	93,80%	-
[38]	Storm-drain and manhole detection	RetinaNet(ResNet50)	-	83,58%	-	-	-	-
		RetinaNet(ResNet101)	-	83,19%	-	-	-	-
		Faster RCNN(ResNet50)	-	92,10%	-	-	-	-
		Faster RCNN(ResNet101)	-	92,35%	-	-	-	-
[39]	Road Pothole	YOLO Darknet CNN	66,50%	-	69,10%	-	-	-
[40]	Indian Pothole Dataset	YOLOv2	52,00%	68,57%	50,00%	-	34,70%	-
		YOLOv3	69,00%	58,79%	35,00%	-	46,29%	-
		YOLOv3 Tiny	76,00%	72,12%	40,00%	-	50,90%	-
[41]	Pothole Image Dataset	YOLOv1	82,00%	79,55%	69,00%	-	-	74,00%
		YOLOv2	81,00%	81,21%	76,00%	-	-	78,00%
		YOLOv3	77,00%	83,60%	78,00%	-	-	78,00%
		YOLOv4	76,00%	80,04%	75,00%	-	-	76,00%
		YOLOv4 Tiny	81,00%	85,48%	83,00%	-	-	82,00%
		YOLOv5	93,00%	95,00%	83,00%	-	-	87,00%
		SSD + MobileNetV2	42,00%	47,40%	56,00%	-	-	47,90%
[42]	Bangladesh Road Pothole	RCNN, SSD, MobileNet	95,00%	-	-	-	-	-
[43]	Cracks Images	YOLOv3	89,16%	-	91,16%	-	-	-
[44]	Malaysia Pothole Dataset	YOLOv5	80,00%	76,00%	73,20%	-	-	-

a high capacity for detecting potholes in various conditions. Similarly, the YOLOv3 model, applied to various datasets, shows great versatility with mAP up to 88.93%, as observed on the “Pavement View Pothole Image Dataset”.

These results highlight the superiority of YOLO models (notably YOLOv8 and YOLOv5) for potholes and other road anomalies, thanks to their high accuracy and high mAP. The combination of a well-annotated data set and advanced models provides optimal performance in road obstacle detection.

2) THE RESULTS OF THE MODELS USED FOR THE CLASSIFICATION OF ROAD OBSTACLES

Classification of road barriers is essential for effective infrastructure management and improved road safety. Classification models distinguish between different types of barriers, providing valuable information for analysis and decision-making. In this section, we will examine the results of models developed for the classification of road barriers. By assessing their performance, we can identify the most effective approaches. The following table 6 presents in a very

detailed way the results obtained for the classification of road obstacles:

Table 6 summarizes the performance of different classification models for Road barriers using multiple data sets. To better visualize the results, we will explore two box-shaped charts (box plots) that present the best performances of the different classification models adopted in the literature. These graphs will allow a clear and concise comparison of model performance, highlighting trends and variations in the results obtained. We will examine two separate box charts that present the most commonly used performance measures for road obstacle classification: accuracy (Accuracy) and recall. The following figures show the accuracy and recall values obtained by the different models applied in the literature.

Analysis of the performance results of different classification models of road obstacles reveals significant variations in terms of accuracy and recall. Emphasis is placed on these two metrics to evaluate the best performance of applied models. Based on Table 6 and Figures 25 and 26, the CNN model applied to the “Smartphone-Pi Pothole Field Capture Dataset + Nienaber Potholes” stands out with an accuracy

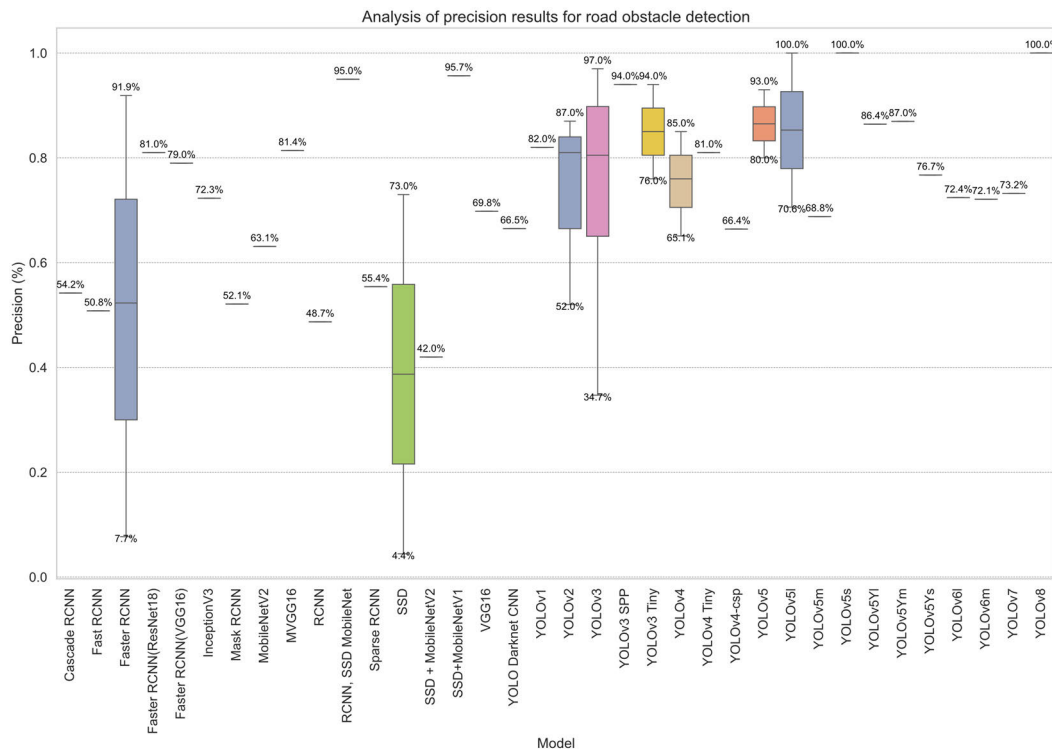


FIGURE 23. Analysis of precision results for road obstacle detection.

(Accuracy) of 99.02% and a recall of 99.03%. These results suggest exceptional performance, offering near-perfect accuracy while maintaining a high recall, indicating that the model is very effective in classifying road barriers while minimizing false negatives. Another notable model is the Random Forest, used on the “Pothole Detection Dataset”, with an accuracy (Accuracy) of 99.70, which indicates that this model is particularly reliable for obstacle classification. In addition, the CNN model applied to the “Road Pothole Variations Image Dataset” has an accuracy (Accuracy) of 99.80% and a recall of 99.60%, providing a remarkable balance between accuracy (Accuracy) and ability to classify the majority of real pothole cases. The ResNet18 and VGG11 models, with an accuracy (Accuracy) of 99.92% and a recall of 99.85% on various data sets, also show outstanding performance. The ResNet50 model on the Bangladesh Road Surface Dataset has an accuracy (Accuracy) of 98.66% and a recall of 99.52%, highlighting its high efficiency in detecting obstacles while capturing most real instances. In addition, the DenseNet121 on the “Annotated Potholes Image Dataset” has an accuracy (Accuracy) of 99.30% and a recall of 99.10%. This model, with results close to those of the best CNN models, proves its effectiveness in the classification of potholes while ensuring a great recall. The best results in terms of accuracy (Accuracy) and recall are mainly from the CNN, ResNet18, VGG11, Random Forest, and DenseNet121 models, which show outstanding performance for road obstacle classification. These models can provide high accuracy (Accuracy)

while maintaining a high recall, which is essential for the reliable classification of road obstacles.

3) THE RESULTS OF MODELS USED FOR ROAD OBSTACLE DETECTION AND CLASSIFICATION

Some models combine road obstacle detection and classification tasks to provide a complete, integrated solution. These models aim to identify the presence of obstacles while accurately categorizing them, which is essential for applications such as autonomous navigation and intelligent infrastructure management. In this section, we will examine the results obtained by these hybrid models, analyzing their performance in both critical aspects of detection and classification. The following table shows these results, giving an overview of the performance of the models studied.

Table 7 summarizes the performance of different detection models and Classification of road barriers using multiple data sets. To better visualize the results, we will explore three box-shaped charts (box plots) that present the best performances of the different detection and classification models adopted in the literature. We will examine three separate box charts that present the most commonly used performance measures for obstacle detection and classification The F1-Score, accuracy, and mAP. The following figures show the F1-Score, precision, and mAP values obtained by different models applied in the literature:

Analysis of the performance results of different detection models and Classification of road obstacles reveals

TABLE 6. Road obstacle classification results.

Refs	Dataset	Model	Precision	mAP	Rappel	Accuracy	IOU	F1-SCORE
[45]	Speed Bump and Pothole Dataset	CNN	98,13%	-	-	-		
[46]	Smartphone-Pi Pothole Field Capture Dataset	CNN	99,02%	98,03%	99,03%	98,33%		
	Nienaber Potholes							
[47]	Chandigarh Thermal Road Image Dataset	CNN	-	-	-	-		
		ResNet101	97,08%	-	-	-		
[48]	Mumbai Pothole Image Dataset	CNN	80,17%	-	-	-		
		ResNet50	85,34%	-	-	-		
		ResNet152	84,48%	-	-	-		
		ResNet50V2	82,76%	-	-	-		
		ResNet152V2	79,31%	-	-	-		
		InceptionV3	85,34%	-	-	-		
		InceptionResNetV2	89,66%	-	-	-		
		DenseNet201	89,66%	-	-	-		
[49]	Muddy and Highway Road Image	ResNet50	94,36%	-	-	-		
		InceptionResNetV2	92,46%	-	-	-		
		VGG19	97,07%	-	-	-		
[50]	Pothole Detection dataset	Random Forest	99,70%	-	-	-		
		AlexNet	86,80%	-	-	-		
		VGG19	87,30%	-	-	-		
		GoogleNet	90,40%	-	-	-		
		ResNet50	93,80%	-	-	-		
[51]	Road Cracks256	CNN	98,00%	-	-	-		
		MobileNetV2	97,23%	-	-	-		
		InceptionV3	90,72%	-	-	-		
		NasNet	32,34%	-	-	-		
[52]	Pothole Image Dataset	ResNet18	99,92%	99,84%	99,85%	99,84%		
	Speed Bump & Hump Dataset	VGG11	99,92%	99,85%	99,85%	99,85%		
	Crack Forest Dataset							
[53]	Potholes and Plain Road Images	CNN	99,92%	99,92%	99,92%	99,92%		
	METU Crack Image Dataset	Filtrage bilateral + Seuillage	98,31%	99,55%	89,52%	94,27%		
[54]	Bituminous Road Pothole Image	CNN + VGG	96,00%	-	-	-		
[55]	Speed Bump Sensor Data	CNN	-	97,00%	97,00%	97,00%		
		LSTM	-	87,50%	92,50%	90,00%		
[16]	iOS Smartphone Accelerometer Dataset	CNN	96,29%	98,00%	84,00%	-		
[56]	Pixabay Road Potholes Images	CNN	92,72%	-	-	-		
	Queensburgh Pothole Image							
	Pothole Detection dataset							
	Pothole600							
[57]	Road Pothole Variations Image Dataset	CNN	99,80%	100%	99,60%	99,60%		
[58]	Real-Time Mobile Pothole Detection Data	CNN	96,00%	-	-	-		
[59]	Concrete Crack Images for Classification	CNN	96,50%	-	-	-		
[60]	KERCD-2023	SVM	98,68%	-	98,61%	98,66%		
		KNN		-	97,21%	97,31%		
		AdaBoost		-	98,11%	98,21%		
		Naive Bayes	95,15%	-	95,67%	95,13%		
		Ensemble Learning	98,68%	-	98,61%	98,66%		
[62]	Pothole Detection dataset	ResNet50	98,05%	-	-	-		

TABLE 6. (Continued.) Road obstacle classification results.

[62]	Bangladesh Road Surface Dataset	ResNet50	98,66%	98,15%	99,52%	99,18%
		VGG16	96,31%	97,69%	97,63%	97,78%
[65]	Annotated Potholes Image Dataset	DenseNet121	99,30%	99,30%	99,10%	99,20%
		VGG19	97,40%	97,40%	98,80%	96,80%
		InceptionV3	96,70%	96,60%	96,30%	96,40%
		InceptionResNetV2	94,80%	94,80%	94,60%	94,20%
		ResNet50	97,60%	97,60%	97,50%	97,64%
[66]	Road Damage Drone Images	VGG19	97,20%	100%	93,00%	96,40%

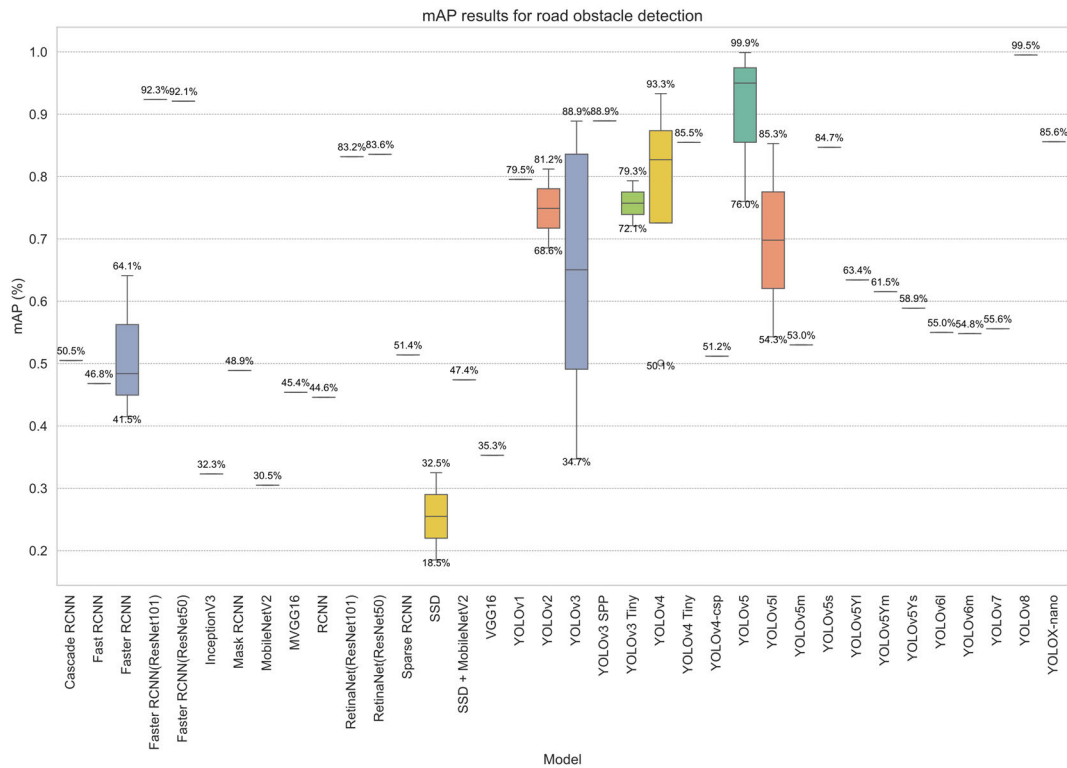


FIGURE 24. mAP results for road obstacle detection.

significant variations in F1-score accuracy and mAP. The emphasis is on these three metrics to evaluate the best performance of applied models. Based on Table 7 and Figures 27, 28, and 29, the CNN model applied to the “Road Defect Image Dataset” shows an accuracy of 93.05% and a F1-score of 90.37%, indicating a good ability to detect and classify obstacles. The improved YOLOv3 Tiny model has a 94% mAP, surpassing the version not improved with an 89% mAP, highlighting the significant impact of Improvements to the model’s performance. For the “Fengtai Street View Image Dataset”, the YOLOv5 model achieves an accuracy of 78.20% and an F1-score of 75%, suggesting moderate performance in detecting obstacles in urban environments. Comparable performance is observed for the YOLOv4 model on the “Taiwan Pavement Defect Image Dataset”, with mAP

of 80.08% and 81.05% for the different versions, indicating a gradual improvement in the performance of the model with architectural modifications. The “Haihe Education Park Road Image Dataset” features a series of YOLO models with slightly varying mAPs: MobileNet+YOLO achieves 96.60% mAP, followed by YOLOv5, YOLOv7, and YOLOv8 with 95.90%, 96.10%, and 96.10% mAPs, respectively. These results indicate the high robustness and reliability of YOLO models for this particular dataset. In comparison, the SSD model shows a mAP of 89.70%, while Faster R-CNN is at 81%, highlighting a lower efficiency compared to the YOLO models for this specific application. The “Road Crack RGB Image Dataset” is another dataset where YOLOv4 shows an accuracy of 80%, a mAP of 80%, and an F1-score of 77%, demonstrating a good overall performance despite

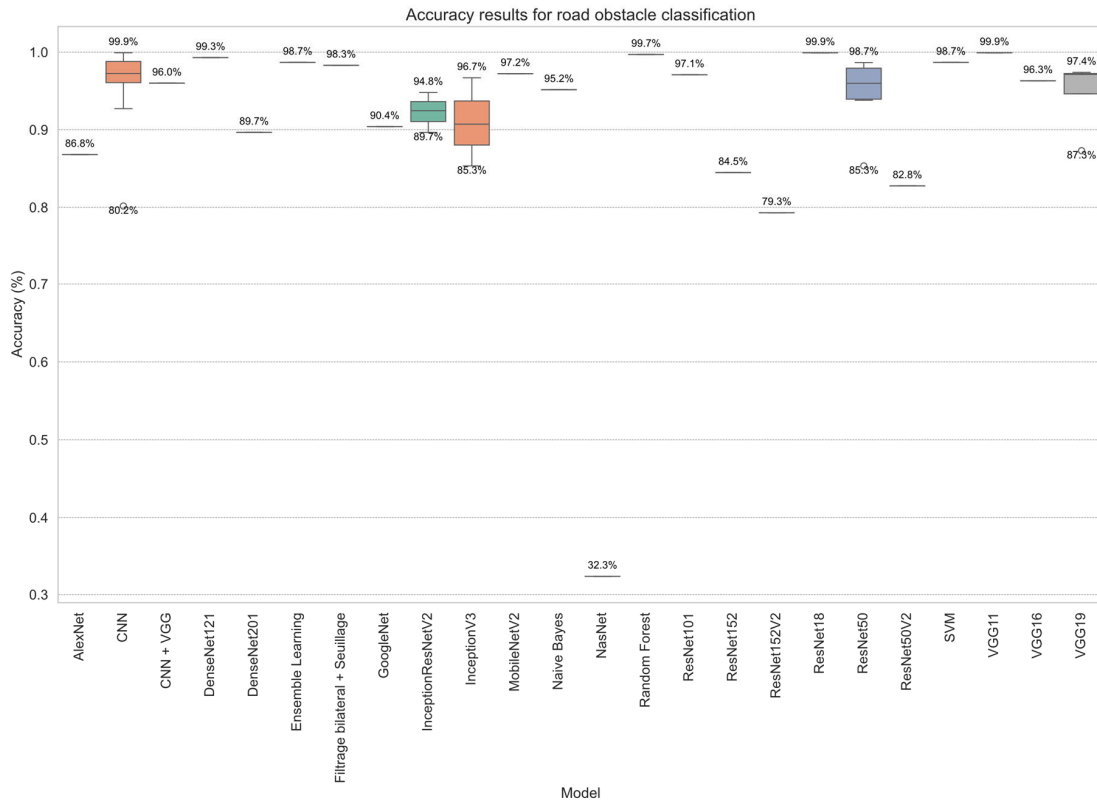


FIGURE 25. Accuracy results for road obstacle classification.

a slight disparity between accuracy and F1-score. In parallel, the «Road Damage Detection 2020» shows more varied results: a model combining YOLOv5-MobileNetV3 with KMeans and other techniques achieves an accuracy of 57.20%, a mAP of 53.60% and an F1-score of 56.60%, indicating modest but balanced performance. On the “Storm-drain and manhole detection” model YOLOv3 Tiny scores impressive with an accuracy of 82%, a mAP of 92%, and an F1-score of 82%, while models YOLOS and Vit show notable variations in terms of mAP and F1-score, which highlights the complexity of detecting roadway overlooks under various conditions. The “Road Cracks256” with a CNN model shows an accuracy of 92%, which is remarkable, and the “Cracks Images Weather Condition” with SSD shows an accuracy of 78.70% and a F1-score of 78.77%, indicating consistency between the two metrics. The performance analysis of different models highlights the importance of the mAP metric in evaluating the effectiveness of road obstacle detection models. YOLO models, in particular, show outstanding performance across multiple data sets, while models like CNN and SSD show varying results depending on the specifics of the input data.

V. SUMMARY

A. DISCUSSION

The purpose of the SLR protocol used in this article is based on the following objectives:

RQ1: The literature review highlights the importance of identifying road obstacles to enhance safety and infrastructure maintenance. Potholes are the most frequently discussed obstacle, appearing in 56 instances, followed by cracks (19), while speed bumps and manhole covers are the least common (6 each). Potholes dominate due to their significant impact on vehicles and safety risks, while cracks also require regular monitoring. The focus of current research is on obstacles with the greatest safety impact, emphasizing the need for advanced detection and management strategies. Future studies should explore under-represented obstacles such as manholes, speed bumps, and dynamic road hazards (e.g., debris or construction zones). There’s also a need for better classification systems that consider context (e.g., urban vs rural impact).

RQ2: The study categorizes research on ML/DL techniques for road obstacles into detection, classification, or combined approaches. Detection, which identifies the presence and location of road features, dominates with 31 studies (44.29%), followed by classification (21 studies, 30%), and combined approaches (18 studies, 25.75%). Detection’s prominence reflects its importance in accurately identifying road features, while classification focuses on organizing and interpreting data. Combined approaches suggest growing maturity in integrating both tasks. A further breakdown (Fig. 6) reveals that potholes are the most studied obstacle, particularly in detection studies (26). Normal roads dominate classification studies (19), and cracks are a key

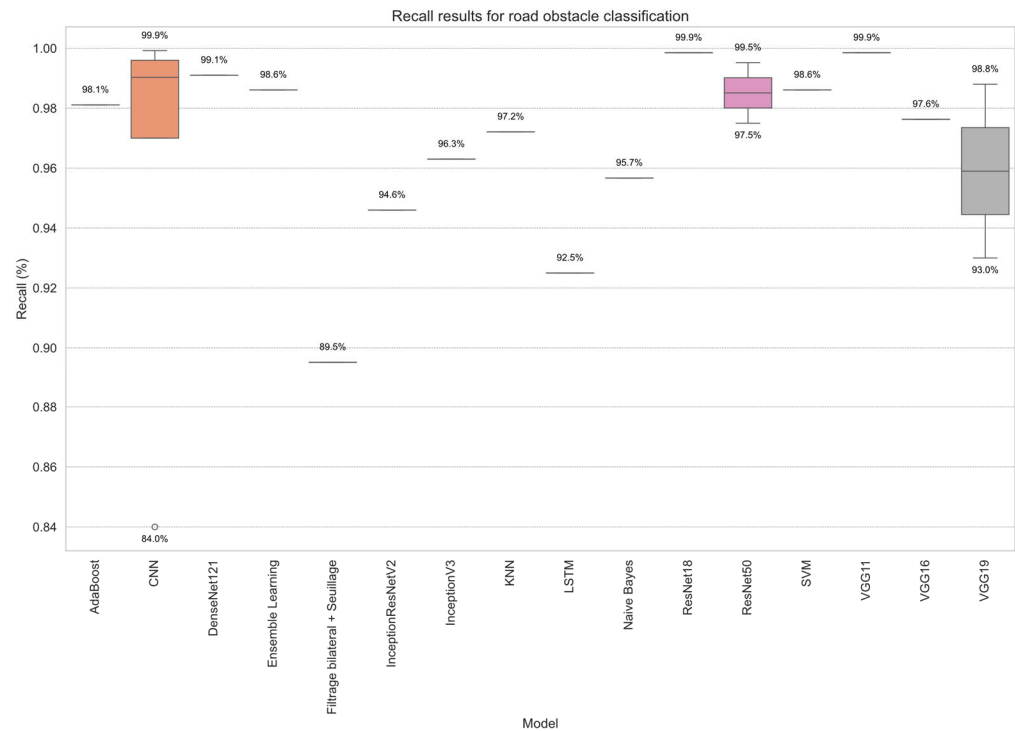


FIGURE 26. Recall results for road obstacle classification.

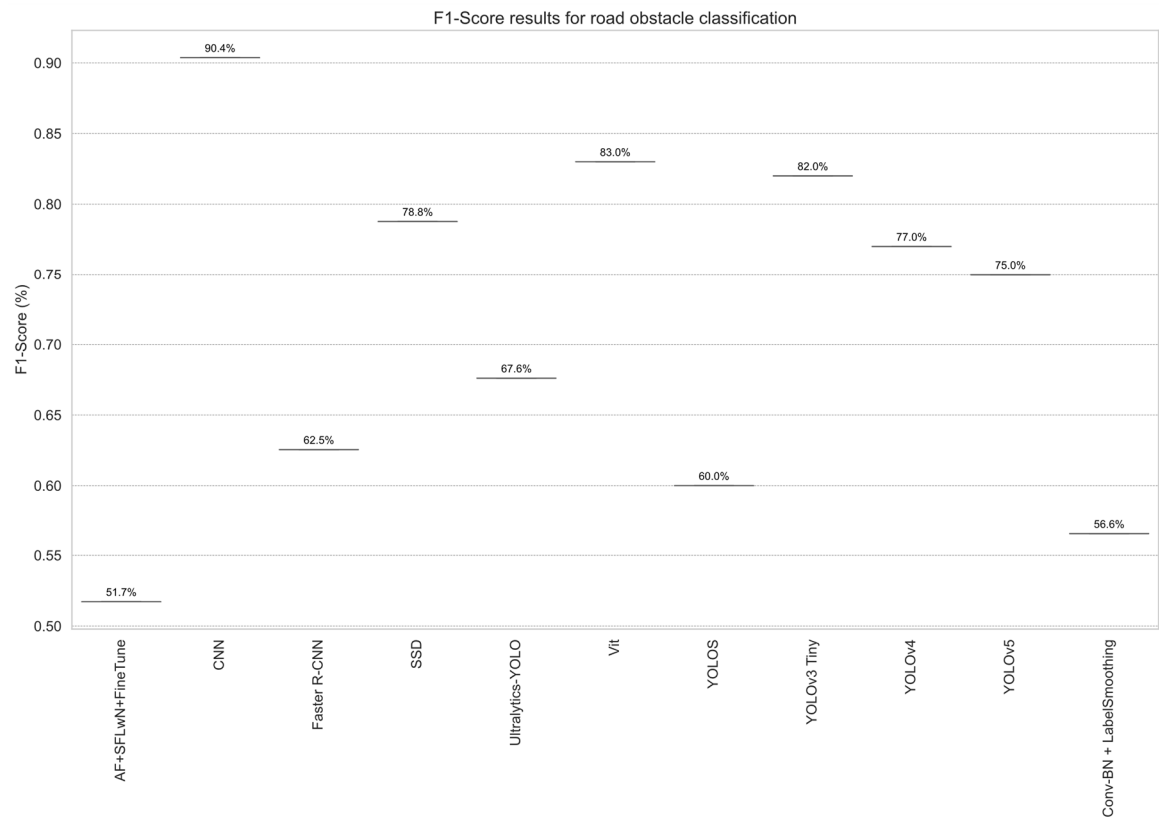


FIGURE 27. F1-score results for road obstacle detection and classification.

focus in combined studies (13). This highlights the field’s emphasis on potholes and the increasing interest in hybrid methodologies for comprehensive analysis. Future research could explore hybrid models that integrate both detection and

TABLE 7. Results obtained for the detection and classification of road obstacles.

Ref s	Dataset	Model	Precision	mAP	Rappel	Accuracy	IOU	F1-SCORE
[67]	Road Condition Image	ResNet50 + YOLOv2	-	-	-	88,90%	-	-
[68]	Road Defect Image	CNN	93,05%	-	87,86%	-	-	90,37%
		YOLOv3 Tiny	-	89,00%	-	89,00%	-	-
		YOLOv3 Tiny	-	94,00%	-	95,00%	-	-
[69]	Fengtai Street View Image Dataset	YOLOv5	78,20%	-	72,10%	-	-	75,00%
[70]	WorldView-3 Satellite Image Dataset	DCNN, Deeplab, Nested-U Net	-	-	-	-	-	-
[71]	Taiwan Pavement Defect Image Dataset	YOLOv3	-	68,06%	-	-	-	-
		YOLOv4-416	-	80,08%	-	-	-	-
		YOLOv4-61	-	81,05%	-	-	-	-
[72]	Haihe Education Park Road Image Dataset	MobileNe t+ YOLO	-	96,60%	-	-	-	-
		YOLOv5	-	95,90%	-	-	-	-
		YOLOv7	-	96,10%	-	-	-	-
		YOLOv8	-	96,10%	-	-	-	-
		SSD	-	89,70%	-	-	-	-
		Faster R-CNN	-	81,00%	-	-	-	-
[73]	Vehicle Motion and GPS Dataset	SVM	-	-	-	88,55%	-	-
		Decision tree	-	-	-	88,35%	-	-
		MLP	-	-	-	79,86%	-	-
[74]	Stellenbosch University Road Surface Image	YOLOv3	-	-	-	82,00%	-	-
		SSD	-	-	-	80,00%	-	-
		HOG + SVM	-	-	-	27,00%	-	-
		Faster R-CNN	-	-	-	74,00%	-	-
[75]	Road Crack RGB Image Dataset	YOLOv4	80,00%	80,00%	77,00%	-	81,00%	77,00%
[76]	Road Damage Detection 2020	YOLOv5- MobileNetV3 + KMeans + CA + Conv- BN + LabelSmoothing"	57,20%	53,60%	56,00%	-	-	56,60%
[77]	Road Damage Detection 2020 China Road Damage Detection	AF+SFLwN+FineTune	-	50,90%	-	-	-	51,72%
[85]	Manhole Cover in Various Conditions Storm-drain and manhole detection	YOLOv3 Tiny	82,00%	92,00%	82,00%	-	-	82,00%
		YOLOS	59,00%	67,00%	77,00%	-	-	60,00%
		Vit	84,00%	91,00%	81,00%	-	-	83,00%
[78]	Dhaka Road Damage Dataset	R-CNN	-	-	-	76,01%	-	-
		Faster R-CNN	-	-	-	99,80%	-	-
[79]	Road Damage Detection 2020	Ultralytics-YOLO	-	-	-	-	-	67,61%
[80]	Road Damage Detection and Classification	Faster R-CNN	-	-	-	-	-	62,55%
[81]	Road Cracks256	CNN	92,00%	-	96,00%	94,00%	-	-
[82]	Rwanda Road Damage	YOLOv5	-	85,00%	-	-	-	-
[83]	Cracks Images Weather Condition	SSD	78,70%	-	79,16%	-	-	78,77%

classification to create more adaptable systems. Additionally, the development of real-time solutions utilizing lightweight deep learning models on edge devices is worth investigating, especially for deployment in autonomous vehicles or smart traffic systems. Furthermore, future work could focus on leveraging large language models (LLMs) to enhance these approaches, opening up new opportunities for advancing road obstacle detection and classification.

RQ3: Datasets are critical for training, validating, and testing ML models in road obstacle detection, ensuring they generalize across scenarios like urban and rural environments. These datasets are divided into public (freely accessible and enabling comparisons) and private (specialized but less widely applicable). Among the 82 datasets analyzed, 52 are private, and 30 are public. Public datasets primarily focus on potholes, followed by cracks, speed

bumps, and manholes. The “Pothole Detection Dataset” and “Annotated Potholes Image Dataset” are the most used, highlighting the dominance of potholes in research. Detection studies often use public datasets for potholes, while classification and combined studies show diverse usage patterns, with datasets like “Road Damage Detection 2020” leading in integrative approaches. Images are the most common data source (91.55%), followed by sensor data (7.04%) and videos (1.41%). This reflects the strong reliance on computer vision techniques. The diversity of obstacle classes includes detailed subclasses for potholes and cracks, while manholes and speed bumps are less studied but categorized clearly. The analysis reveals gaps in datasets and highlights the need for balanced research across obstacle types. New public datasets with diverse road conditions and obstacle types (e.g., seasonal variations, lighting conditions) are needed. Crowd-sourced

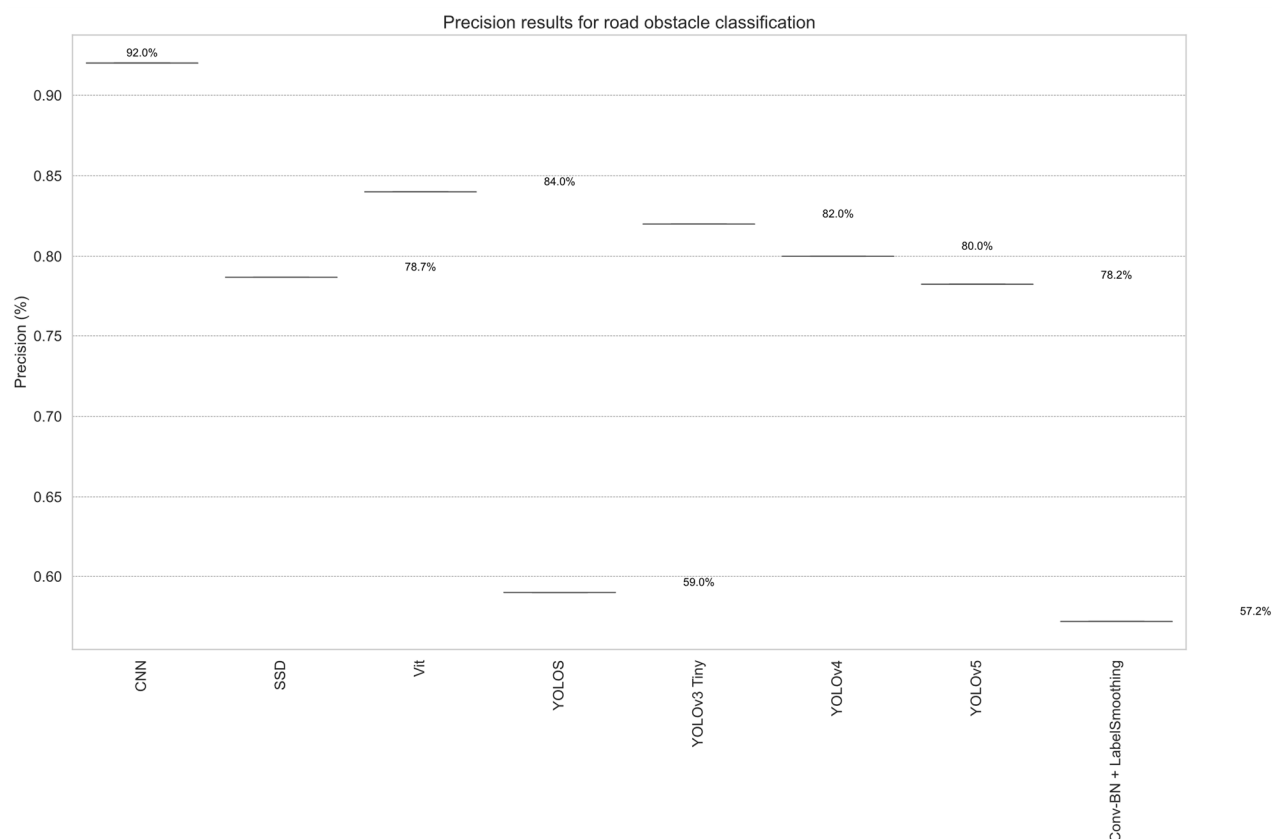


FIGURE 28. Precision results for road obstacle detection and classification.

and synthetic data generation (e.g., via GANs) could help fill data gaps. One notable limitation identified in the literature is the heavy reliance on private datasets, which accounted for 52 of the 95 studies. This restricts reproducibility, limits benchmarking opportunities, and hinders progress toward generalizable solutions. We recommend that future studies prioritize the use and creation of public datasets to promote transparency and comparability across research efforts.

RQ4: Pre-processing is a critical step in preparing datasets for road obstacle detection, improving data quality, and enhancing model performance. Techniques like data augmentation (30.86%), annotation (24.69%), and resizing (17.28%) dominate the literature due to their effectiveness in diversifying datasets, standardizing image dimensions, and improving labeling for object detection. Data augmentation, in particular, employs methods such as rotation, flipping, and scaling to enhance data diversity and model generalization. Less common techniques like noise reduction, cropping, normalization, and data filtering address specific data quality issues, while specialized methods such as geometric correction and class balancing cater to targeted needs. Dataset-specific pre-processing reveals that public datasets, such as the “Pothole Detection Dataset” and “Annotated Potholes Image Dataset,” frequently utilize augmentation and resizing to optimize data for analysis. These techniques collectively play

a vital role in enabling robust and accurate road obstacle detection models. Advanced augmentation techniques such as domain adaptation and transfer learning should be investigated. Semi-automated or AI-assisted annotation tools can also reduce dataset preparation time.

RQ5: Deep Learning dominates methodological approaches in road obstacle detection, with 87.14% of studies relying solely on it due to its superior capabilities in object detection and classification. Among deep learning models, YOLO is the most popular, particularly YOLOv3, YOLOv5, and YOLOv4, highlighting its efficiency and accuracy. CNNs follow, valued for their ability to extract complex features, while other models like Faster R-CNN, SSD, and ResNet offer balanced trade-offs between speed and precision. Machine Learning alone is rarely used (2.86%), with models like SVM, KNN, and Random Forest appearing in isolated cases. Hybrid approaches combining Deep Learning and Machine Learning (10%) are explored but remain uncommon due to integration complexity. This trend underscores the dominance of Deep Learning, particularly YOLO and CNN, in handling complex detection tasks, while other methods are applied selectively for specific needs, reflecting a dynamic exploration of diverse approaches.

Based on our review results, YOLO models (notably YOLOv5 and YOLOv8) demonstrate outstanding detection

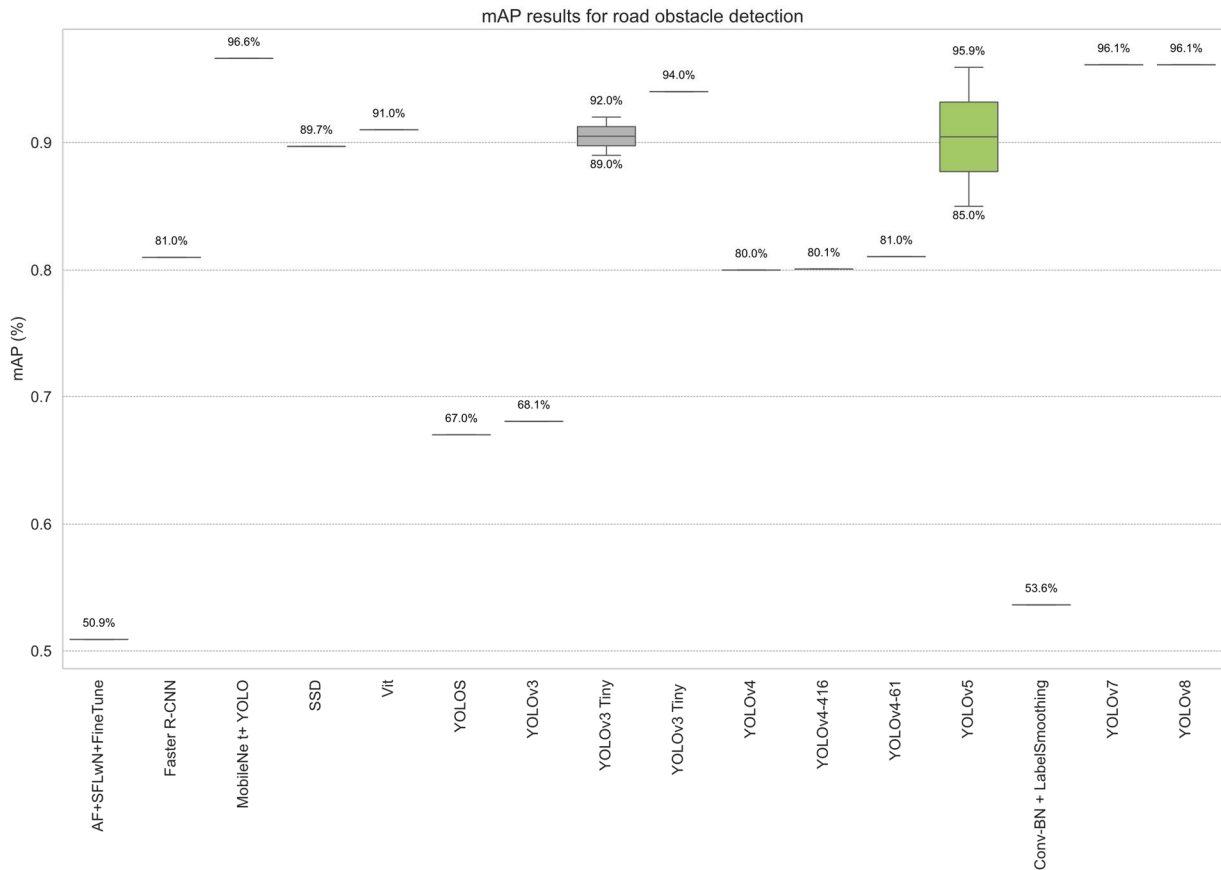


FIGURE 29. mAP results for road obstacle detection and classification.

performance, achieving high accuracy and mean Average Precision (mAP) values (e.g., up to 100% accuracy and 99.5% mAP on the “Annotated Potholes Image Dataset”). Their fast inference speed and lower computational requirements make them well-suited for real-time applications such as autonomous driving and edge deployments.

In contrast, CNN-based models like ResNet18, DenseNet121, and VGG11 excel in classification tasks, showing near-perfect accuracy and recall on datasets such as the “Smartphone-Pi Pothole Field Capture Dataset.” While offering strong feature extraction, CNNs typically require more computational resources and longer processing times, limiting real-time usability.

Hybrid models combining detection and classification, such as improved YOLOv3 Tiny and MobileNet+YOLO, show balanced performance across accuracy and speed, offering promising solutions where both are important.

Thus, the choice between YOLO and CNN depends on application needs: YOLO for efficient, real-time detection, CNN for detailed classification, and hybrids for integrated tasks.

Integration of multi-sensor fusion (e.g., combining LiDAR, radar, and vision) can enhance detection robustness. Exploring transformer-based models (e.g., DETR) and

efficient architectures like YOLO-NAS could offer performance gains.

RQ6: Evaluating the performance of models is crucial to ensuring their reliability and effectiveness. Researchers utilize various metrics to quantify precision, sensitivity, specificity, and other aspects, enabling the comparison of model results and the identification of best practices. Performance measures such as precision (23.84%), accuracy (23.84%), recall (19.87%), mAP (14.57%), and F1-score (14.57%) are most commonly used, with IoU being the least frequent (3.31%). The emphasis on precision and accuracy highlights their importance in correctly identifying obstacles and ranking predictions. Recall measures the ability to detect relevant obstacles, while mAP and F1-score, though slightly less prevalent, evaluate average performance and the balance between precision and recall. IoU is often reserved for advanced or specialized contexts. Future work should standardize evaluation metrics and thresholds for comparison across studies. New metrics could consider latency, robustness under weather/noise, or energy efficiency.

RQ7: The results of the systematic literature review reveal the consistent superiority of YOLO models, particularly YOLOv8 and YOLOv5, in road obstacle detection, with these models achieving exceptionally high mAP and accuracy

across diverse datasets. For instance, YOLOv8 demonstrated outstanding performance on the “Annotated Potholes Image Dataset,” achieving 100% accuracy and a mAP of 99.50%, while YOLOv5 achieved a near-perfect mAP of 99.90% on the “Road Conditions and Potholes Dataset.” Similarly, YOLOv4 and YOLOv3 displayed strong results on datasets like the “Diverse Pothole Road Image Dataset” and the “Pavement View Pothole Image Dataset,” showcasing their versatility and adaptability. For classification tasks, CNN-based models such as ResNet18, DenseNet121, VGG11, and Random Forest consistently excelled, achieving near-perfect accuracy and recall values. Notable examples include the CNN model achieving 99.02% accuracy and 99.03% recall on the “Smartphone-Pi Pothole Field Capture Dataset,” and the DenseNet121 model achieving 99.30% accuracy and 99.10% recall on the “Annotated Potholes Image Dataset.” Models that integrate detection and classification tasks, such as improved YOLOv3 Tiny and MobileNet+YOLO, demonstrated balanced and robust performance across F1-score, mAP, and accuracy metrics, achieving mAP values as high as 96.60% on the “Haihe Education Park Road Image Dataset.” These findings underscore the critical role of selecting performance metrics tailored to study objectives, as mAP and precision are vital for detection tasks. In contrast, classification tasks benefit more from high accuracy and recall. Overall, the review highlights the effectiveness of advanced architectures like YOLO and CNNs, particularly when paired with well-annotated datasets, in achieving state-of-the-art performance for road obstacle detection, classification, and integrated tasks, paving the way for reliable and efficient automated driving systems. Future research could benchmark models across diverse scenarios (e.g., weather, night-time) and explore continual learning approaches to adapt models post-deployment.

In general, we have synthesized findings from various studies on road obstacle detection and classification, identifying key external factors influencing these systems’ performance. These factors include road conditions, weather, traffic, and the presence of pedestrians, vehicles, motorcycles, and cyclists, which significantly affect the ability of models to detect and classify obstacles accurately. To improve the effectiveness of these systems, it is crucial to develop an integrated approach that classifies obstacles according to their type and environmental context. The application of real-time obstacle detection and classification in autonomous vehicles has several implications for road safety. First, it can enhance in-vehicle systems by alerting the driver to obstacles or road anomalies in real time, improving situational awareness and helping to prevent accidents. Secondly, such systems can support law enforcement by identifying vehicles navigating hazardous areas or exhibiting abnormal behavior, allowing authorities to intervene before accidents occur. These systems can detect potentially unsafe driving patterns by analyzing factors such as vehicle speed, distance, and environmental conditions. Moreover, road safety can be improved for infrastructure planning by identifying high-risk areas based

on frequent obstacle detection failures or hazardous conditions. This type of system can also assist fleet operators in monitoring driver behavior and predicting future risk factors, leading to more informed hiring and operational decisions. Together, these applications not only improve road safety but also contribute to reducing repair costs and preventing accidents by proactively addressing potential hazards.

B. IMPACT AND REAL-WORLD IMPLICATIONS

The findings from this systematic literature review reveal several real-world implications of road obstacle detection systems powered by ML and DL, particularly in the domains of social well-being, environmental sustainability, and governmental efficiency.

- **Social Impact:** The social implications of automated road obstacle detection are considerable. By enabling real-time identification of hazardous road conditions, such as potholes, cracks, and other anomalies, these technologies improve overall road safety for all users, including drivers, cyclists, motorcyclists, and pedestrians, especially disabled or elderly people are more affected by poor road conditions/obstacles. This is particularly important in areas where infrastructure maintenance is inconsistent or delayed, disproportionately affecting low-income and rural communities. In addition, the integration of these systems into Advanced Driver Assistance Systems (ADAS) and Autonomous Vehicles (AV) increases driver awareness and reduces the risk of accidents. The ability to detect and react to road anomalies in real time contributes to safer mobility, reduces injury rates, and improves public health outcomes. Ultimately, these advances contribute to more inclusive and equitable access to transportation systems.
- **Environmental Impact:** From an environmental point of view, effective detection of road anomalies and timely maintenance reduce vehicle damage and fuel-related inefficiencies. Poor road conditions often increase fuel consumption due to sudden stops, detours, or irregular driving patterns. Maintaining smoother roads can therefore help reduce greenhouse gas emissions and energy consumption. In addition, by preventing the escalation of minor road damage in major reconstruction projects, these systems help to reduce the frequency and intensity of roadworks, which are a source of air and noise pollution. In the long term, better-managed road networks support the goals of sustainable transport and climate-resilient infrastructure planning.
- **Governmental Impact:** Government implications are equally important. Road monitoring tools based on ML and DL provide public authorities with objective, data-driven information about the state of roads in large geographic areas. This allows for better prioritization of maintenance tasks, optimized allocation of limited budgets, and proactive infrastructure management. In addition, municipalities can use these systems for

accountability and transparency by documenting how and where public funds are used to maintain infrastructure. This information can also help enforce road safety rules, identify areas of recurring risk, and plan urban development with a focus on mobility equity and public safety. In addition, the use of AI for road surveillance aligns with broader government strategies for digital transformation and smart city development, enabling more efficient and responsive governance. Moreover, governments can accelerate the deployment of these intelligent monitoring systems by fostering public-private partnerships, leveraging the innovation and agility of the private sector while ensuring alignment with public interests.

C. CHALLENGES AND PERSPECTIVES

Despite notable progress, the field faces several challenges and future directions: A key concern is the generalizability of the model, as many current approaches struggle with extreme or variable conditions such as night driving, rain, fog, or snow, which are common in real-world scenarios. Data-related problems persist, including limited public datasets, lack of diversity in road types and obstacle categories, and a serious imbalance between classes, with potholes vastly over-represented compared to less-studied but equally relevant obstacles such as road bumps, Manholes, or temporary hazards. Manual annotation remains time-consuming and costly, highlighting the need for semi-automated or AI-assisted labeling methods to streamline dataset preparation. What's more, most existing models do not incorporate contextual information, such as urban and rural environments, road geometry, or traffic density, which could significantly improve the accuracy and robustness of detection. The potential of large language models (LLMs) is also underutilized; LLMs could contribute to automatic annotation, explainable AI, and semantic metadata generation. Real-time deployment is another challenge, requiring a lightweight, energy-efficient architecture that can run on edge devices, such as those used in autonomous vehicles or intelligent infrastructure systems. Addressing these limitations is essential to developing more robust, scalable, and context-aware road safety systems.

VI. CONCLUSION

Our paper presented an in-depth analysis of existing literature on detecting and classifying road obstacles, using a rigorous and structured methodology based on Systematic Literature Review (SLR). By following the PRISMA guidelines, we were able to identify, select, and critically review relevant studies, ensuring a comprehensive and unbiased assessment of available knowledge.

This analysis highlighted the types of road obstacles dealt with, the types of studies addressed, the data sets and sources used, the pre-processing techniques applied, the models and techniques employed, the performance measures used, as well as the results obtained in the articles reviewed. Consequently, this state-of-the-art review provides a clear and

detailed understanding of current advances and challenges in the field of obstacle detection and classification, highlighting key trends and research gaps, and underscoring the need to develop innovative technologies and strategies to improve road infrastructure safety and maintenance.

Among the models reviewed, YOLOv8 and YOLOv5 emerged as top performers for real-time obstacle detection, demonstrating high accuracy and mean Average Precision (mAP) across multiple public datasets. Their efficiency and speed make them particularly suitable for deployment in autonomous vehicles and real-time traffic monitoring systems. For detailed classification tasks, CNN-based models such as ResNet18 and DenseNet121 showed superior accuracy and recall, which may be more appropriate for offline or high-precision analysis. We recommend that practitioners select models aligned with their application goals: YOLO variants for real-time detection needs and CNN architectures for classification-focused tasks. Future work should explore hybrid models combining the strengths of both approaches to enhance robustness and applicability in complex road environments.

While this review focused on computer vision-based approaches for road obstacle detection and classification, future research could benefit from a broader, interdisciplinary perspective. Emerging directions include the integration of multi-sensor fusion (e.g., combining LiDAR, radar, and vision), which can enhance obstacle detection in complex environments. Additionally, machine learning-based forecasting techniques can help predict obstacle occurrences and support proactive road maintenance. The incorporation of crowdsourced data, such as user-reported hazards via mobile apps, can enrich training datasets and increase system robustness. Lastly, aligning these technologies with smart city infrastructure (e.g., IoT-connected roadways, real-time traffic management systems) can support scalable, real-time implementation. These avenues offer promising directions for future research beyond the current scope of computer vision.

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